Schur positivity and the nabla operator

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CART Seminar

Main results

Let $\mu, \nu, \lambda \vdash n$ and $r \in \mathbb{Z}_{\geq 1}$.

Theorem (Qiu–Zhang)

$$(-1)^{n-\ell(\mu)} \langle \nabla^r m_\mu, s_\lambda \rangle \in \mathbb{N}[q, t].$$

Qiu–Zhang (2026a).

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$$(-1)^{n-\ell(\mu)} \langle \nabla^r m_\mu, s_\lambda \rangle \in \mathbb{N}[q,t].$$

Theorem (Gu–Kang–Qiu–Zhang)

If $\mu \trianglerighteq \nu$, then

$$\left\langle \nabla^r \omega \widetilde{H}_\mu[X;0,t^{-1}] - \nabla^r \omega \widetilde{H}_\nu[X;0,t^{-1}],\ s_\lambda \right\rangle \in \mathbb{N}[q,t].$$

$r = 1$: conjectures of Bergeron–Garsia–Haiman–Tesler.

Qiu–Zhang (2026a). Gu–Kang–Qiu–Zhang (2026).

Symmetric functions and Schur positivity

Degree-n bases of the symmetric function ring $\Lambda_{\mathbb{Q}(q,t)}$:

$$\{m_\mu : \mu \vdash n\}, \qquad \{s_\lambda : \lambda \vdash n\}.$$

m_μ: the sum of distinct monomials with exponent partition μ.

$$m_{(2,1)} = \sum_{i \neq j} x_i^2 x_j = s_{(2,1)} - 2s_{(1,1,1)}.$$

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Schur positivity

$f = \sum_\lambda a_\lambda(q,t) s_\lambda$ is *Schur positive* if

$$a_\lambda(q,t) = \langle f, s_\lambda \rangle \in \mathbb{N}[q,t] \qquad \text{for every } \lambda.$$

Hall inner product: $\langle s_\lambda, s_\mu \rangle = \delta_{\lambda\mu}$.

Macdonald polynomials

Macdonald scalar product

$$\langle p_\lambda, p_\mu \rangle_{q,t} = \delta_{\lambda\mu}\, z_\lambda \prod_i \frac{1-q^{\lambda_i}}{1-t^{\lambda_i}}, \qquad z_\lambda = \prod_{j\geq 1} j^{m_j(\lambda)} m_j(\lambda)!.$$

Triangularity and orthogonality

$$P_\mu = m_\mu + \sum_{\lambda \lhd \mu} u_{\lambda\mu}(q,t) m_\lambda, \qquad \langle P_\mu, P_\nu \rangle_{q,t} = 0 \quad (\mu \neq \nu).$$

$\lhd$: strict dominance order; $u_{\lambda\mu} \in \mathbb{Q}(q,t)$.

Macdonald, Symmetric Functions and Hall Polynomials, 2nd ed., Oxford, 1995, Chapter VI.

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Integral form

$$J_\mu = c_\mu P_\mu \in \Lambda_{\mathbb{Z}[q,t]}, \qquad c_\mu = \prod_{c\in\mu} (1 - q^{a(c)} t^{\ell(c)+1}).$$

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The modified Macdonald basis

$$\widetilde{H}_\mu[X;q,t] = t^{n(\mu)} J_\mu\left[\frac{X}{1-t^{-1}};q,t^{-1}\right], \qquad n(\mu) = \sum_{i\geq 1}(i-1)\mu_i.$$

$$\widetilde{H}_\mu[X;q,t] = \sum_{\lambda\vdash|\mu|} \widetilde{K}_{\lambda\mu}(q,t)s_\lambda.$$

Macdonald positivity

Haiman proved $\widetilde{K}_{\lambda\mu}(q,t) \in \mathbb{N}[q,t]$.

Haiman, J. Amer. Math. Soc. 14 (2001), 941–1006.

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$$\underbrace{P_\mu}_{\text{monic and orthogonal}} \longrightarrow \underbrace{J_\mu}_{\text{integral}} \longrightarrow \underbrace{\widetilde{H}_\mu}_{\text{Schur-positive}} .$$

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The nabla operator and its eigenvalues

$$\nabla \widetilde{H}_\mu[X;q,t] = T_\mu \widetilde{H}_\mu[X;q,t], \qquad T_\mu = \prod_{c\in\mu} q^{a'(c)} t^{\ell'(c)} = q^{n(\mu')} t^{n(\mu)}.$$

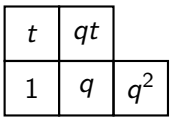
French notation: column j, row i gives $a'(c) = j-1$, $\ell'(c) = i-1$.

Bergeron–Garsia, CRM Proc. Lecture Notes 22 (1999), 1–52.

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$$\mu = (3,2), \qquad T_{(3,2)} = (1)(q)(q^2)(t)(qt) = q^4t^2.$$

Degree 3

$$T_{(3)} = q^3, \qquad T_{(2,1)} = qt, \qquad T_{(1,1,1)} = t^3.$$

Bergeron–Garsia, CRM Proc. Lecture Notes 22 (1999), 1–52.

Example: changing basis for $m_{(2,1)}$

Degree 3

$$\widetilde{H}_{(3)} = s_{(3)} + (q+q^2)s_{(2,1)} + q^3 s_{(1,1,1)},$$
$$\widetilde{H}_{(2,1)} = s_{(3)} + (q+t)s_{(2,1)} + qt\, s_{(1,1,1)},$$
$$\widetilde{H}_{(1,1,1)} = s_{(3)} + (t+t^2)s_{(2,1)} + t^3 s_{(1,1,1)}.$$

$$m_{(2,1)} = s_{(2,1)} - 2s_{(1,1,1)}, \qquad R = q^2 + qt + t^2 + 2q + 2t + 2.$$

$$m_{(2,1)} = -\frac{t+2}{(q-t)(q^2-t)}\,\widetilde{H}_{(3)} + \frac{R}{(q^2-t)(q-t^2)}\,\widetilde{H}_{(2,1)} - \frac{q+2}{(q-t)(q-t^2)}\,\widetilde{H}_{(1,1,1)}.$$

Example: applying ∇ to $m_{(2,1)}$

For $\mu = (2,1)$, the sign is $(-1)^{3-2} = -1$.

$$\nabla m_{(2,1)} = -2s_{(3)} - A(q,t)s_{(2,1)} - B(q,t)s_{(1,1,1)},$$

where

$$A(q,t) = qt(q+t+2) + 2(q^2+q+t^2+t),$$
$$B(q,t) = qt(q^2+qt+t^2) + 2(q^3+q^2t+qt^2+qt+t^3).$$

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Theorem (Qiu–Zhang)

For $\mu, \lambda \vdash n$ and $r \in \mathbb{Z}_{\geq 1}$, we have

$$(-1)^{n-\ell(\mu)}\langle \nabla^r m_\mu, s_\lambda \rangle \in \mathbb{N}[q,t].$$

Qiu–Zhang (2026a), Theorem 1.2.

Modified Hall–Littlewood polynomials

Specialization at $q = 0$

$$\widetilde{H}_\mu[X; 0, t].$$

Degree 3

$$\begin{aligned}
\widetilde{H}_{(3)}[X; 0, t] &= s_{(3)}, \\
\widetilde{H}_{(2,1)}[X; 0, t] &= s_{(3)} + t\, s_{(2,1)}, \\
\widetilde{H}_{(1,1,1)}[X; 0, t] &= s_{(3)} + (t + t^2) s_{(2,1)} + t^3 s_{(1,1,1)}.
\end{aligned}$$

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Dominance positivity

$$\omega \widetilde{H}_\mu[X; 0, t^{-1}], \qquad \omega s_\lambda = s_{\lambda'}.$$

$\nabla = \nabla_{q,t}$ retains both parameters.

Example: changing basis for a Hall–Littlewood difference

Dominance: $\mu \trianglerighteq \nu \iff \sum_{i\le k}\mu_i \ge \sum_{i\le k}\nu_i$ for every k. Cover: $(2,1) \gtrdot (1,1,1)$, with no intervening partition.

$$\omega\widetilde{H}_{(2,1)}[X;0,t^{-1}] = s_{(1,1,1)} + t^{-1}s_{(2,1)},$$
$$\omega\widetilde{H}_{(1,1,1)}[X;0,t^{-1}] = s_{(1,1,1)} + (t^{-1}+t^{-2})s_{(2,1)} + t^{-3}s_{(3)}.$$

$$\omega\widetilde{H}_{(2,1)}[X;0,t^{-1}] - \omega\widetilde{H}_{(1,1,1)}[X;0,t^{-1}] = -t^{-3}s_{(3)} - t^{-2}s_{(2,1)}.$$

Modified Macdonald expansion

$$\omega\widetilde{H}_{(2,1)}[X;0,t^{-1}] - \omega\widetilde{H}_{(1,1,1)}[X;0,t^{-1}] = \frac{\widetilde{H}_{(2,1)}}{t(q-t^2)} - \frac{q\,\widetilde{H}_{(1,1,1)}}{t^3(q-t^2)}.$$

Example: applying ∇ to the Hall–Littlewood difference

Apply ∇

$$\nabla\omega\big(\widetilde{H}_{(2,1)}[X;0,t^{-1}] - \widetilde{H}_{(1,1,1)}[X;0,t^{-1}]\big) = \frac{q}{q-t^2}\big(\widetilde{H}_{(2,1)} - \widetilde{H}_{(1,1,1)}\big).$$

$$\widetilde{H}_{(2,1)} - \widetilde{H}_{(1,1,1)} = (q-t^2)\big(s_{(2,1)} + t\, s_{(1,1,1)}\big).$$

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Individual images

$$\nabla\omega\widetilde{H}_{(2,1)}[X;0,t^{-1}] = s_{(3)} + (q+t+t^2)s_{(2,1)} + (qt+t^3)s_{(1,1,1)},$$
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$$\nabla\omega\big(\widetilde{H}_{(2,1)}[X;0,t^{-1}]-\widetilde{H}_{(1,1,1)}[X;0,t^{-1}]\big)=\frac{q}{q-t^2}\big(\widetilde{H}_{(2,1)}-\widetilde{H}_{(1,1,1)}\big).$$

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Theorem (Gu–Kang–Qiu–Zhang)

For $\mu,\nu,\lambda\vdash n$ with $\mu\trianglerighteq\nu$ and $r\in\mathbb{Z}_{\geq 1}$, we have

$$\big\langle\nabla^r\big(\omega\widetilde{H}_\mu[X;0,t^{-1}]-\omega\widetilde{H}_\nu[X;0,t^{-1}]\big),s_\lambda\big\rangle\in\mathbb{N}[q,t].$$

Gu–Kang–Qiu–Zhang (2026), Corollary 5.12.

Previous work: monomial symmetric functions

- $\mu = (1^n)$: Haiman—diagonal harmonics.

$$\mathrm{Frob}_{q,t}(DH_n) = \nabla e_n, \qquad e_n = m_{(1^n)}.$$

Haiman, Invent. Math. 149 (2002).

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- $\mu = (2^k, 1^\ell)$: Qu–Xin's positive C-operator recursion.

$$\text{signed monomial} \quad \longrightarrow \quad \text{positive expansion in } C_\alpha(1).$$

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Previous work: modified Hall–Littlewood polynomials

Theorem (Haglund–Morse–Zabrocki; Carlsson–Mellit)

For $\mu, \lambda \vdash n$, we have

$$(-1)^{n-\ell(\mu)} \langle \nabla \widetilde{H}_\mu[X; 0, t], s_\lambda \rangle \in \mathbb{N}[q, t].$$

HMZ, Canad. J. Math. 64 (2012); Carlsson–Mellit, J. Amer. Math. Soc. 31 (2018); Grojnowski–Haiman (2007).

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Theorem (Qu, 2026)

For $\mu, \nu \vdash n$ with at most two columns, $\mu \trianglerighteq \nu$, $r \geq 1$, and $\lambda \vdash n$, we have

$$\left\langle \nabla^r \omega \widetilde{H}_\mu[X; 0, t^{-1}], s_\lambda \right\rangle \in \mathbb{N}[q, t],$$

$$\left\langle \nabla^r \omega \widetilde{H}_\mu[X; 0, t^{-1}] - \nabla^r \omega \widetilde{H}_\nu[X; 0, t^{-1}],\ s_\lambda \right\rangle \in \mathbb{N}[q, t].$$

HMZ, Canad. J. Math. 64 (2012); Carlsson–Mellit, J. Amer. Math. Soc. 31 (2018); Grojnowski–Haiman (2007). Qu (2026), Corollaries 6.1–6.2.

The C-operators

For $a \geq 1$, Haglund–Morse–Zabrocki define

$$C_a(f)[X] = \left(-\frac{1}{q}\right)^{a-1} \langle z^a \rangle f\left[X - \frac{1-1/q}{z}\right] \sum_{r \geq 0} h_r[X] z^r.$$

For a composition $\alpha = (\alpha_1, \ldots, \alpha_m)$,

$$C_\alpha(f) := C_{\alpha_1}(C_{\alpha_2}(\cdots C_{\alpha_m}(f)\cdots)).$$

Haglund–Morse–Zabrocki, Canad. J. Math. 64 (2012), 822–844.

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$$e_n = \sum_{\alpha \vDash n} C_\alpha(1).$$

Haglund–Morse–Zabrocki, Canad. J. Math. 64 (2012), 822–844.

Schur positivity from the shuffle theorem

For $\alpha \vDash n$, $\lambda \vdash n$, and $r \geq 1$, we have

$$\left\langle \nabla^r C_\alpha(1), s_\lambda \right\rangle \in \mathbb{N}[q,t].$$

Carlsson–Mellit, J. Amer. Math. Soc. 31 (2018); Mellit, Duke Math. J. 170 (2021); Grojnowski–Haiman, unpublished manuscript (2007).

Schur positivity from the shuffle theorem

For $\alpha \vDash n$, $\lambda \vdash n$, and $r \geq 1$, we have

$$\left\langle \nabla^r C_\alpha(1), s_\lambda \right\rangle \in \mathbb{N}[q,t].$$

$$\nabla^r C_\alpha(1) \longrightarrow \text{nonnegative sum of column LLT polynomials} \longrightarrow \text{Schur-positive.}$$

- $r = 1$: the compositional shuffle theorem.
- $r \geq 1$: Mellit's compositional (km, kn)-shuffle theorem.

Carlsson–Mellit, J. Amer. Math. Soc. 31 (2018); Mellit, Duke Math. J. 170 (2021); Grojnowski–Haiman, unpublished manuscript (2007).

Proof strategy

Monomial symmetric functions

$$(-1)^{n-\ell(\mu)} m_\mu \longrightarrow \text{positive } C\text{-expansion} \xrightarrow{\text{the shuffle theorem}} \text{Schur positivity.}$$

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Hall–Littlewood dominance differences

For $\mu \trianglerighteq \nu$, choose $d \in \mathbb{Z}$ and $\boldsymbol{\sigma}$ with

$$\omega \widetilde{H}_\mu[X; 0, t^{-1}] - \omega \widetilde{H}_\nu[X; 0, t^{-1}] = -t^d \, \mathrm{LLT}_{\boldsymbol{\sigma}}[X; t].$$

For $r \geq 1$, we obtain

$$-t^d \nabla^r \, \mathrm{LLT}_{\boldsymbol{\sigma}}[X; t] = \sum_{\boldsymbol{\tau}} c_{\boldsymbol{\tau}}(q, t) \, \mathrm{LLT}_{\boldsymbol{\tau}}[X; t], \qquad c_{\boldsymbol{\tau}}(q, t) \in \mathbb{N}[q, t^{\pm 1}].$$

Schur coefficients lie in $\mathbb{N}[q, t^{\pm 1}]$; polynomiality gives $\mathbb{N}[q, t]$.

General dominance: sum along a chain of covers.

A recursion with nonnegative coefficients

For nonempty partitions, write $\nu \prec \mu$ if

$$\ell(\nu) < \ell(\mu), \qquad \text{or} \qquad \ell(\nu) = \ell(\mu) \text{ and } \min \nu < \min \mu.$$

$\varnothing \prec \mu$ for $\mu \neq \varnothing$.

Theorem A (Qiu–Zhang)

For nonempty $\mu \vdash n$,

$$(-1)^{n-\ell(\mu)} m_\mu = \sum_{j=1}^{n} \sum_{\substack{\nu \vdash n-j \\ \nu \prec \mu}} b_{\mu;j,\nu}(q)\, C_j\Big((-1)^{|\nu|-\ell(\nu)} m_\nu\Big),$$

with explicit $b_{\mu;j,\nu}(q) \in \mathbb{Q}_{\geq 0}[q]$.

Qiu–Zhang (2026a), Theorem 4.1; coefficients normalized by the part-multiplicity factorials.

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Initial value: $m_\varnothing = 1 = C_\varnothing(1)$.

Qiu–Zhang (2026a), Theorem 4.1; coefficients normalized by the part-multiplicity factorials.

Iterating **Theorem A** gives a positive C-expansion

For $\nu \prec \mu$, assume inductively that

$$(-1)^{|\nu|-\ell(\nu)} m_\nu = \sum_{\beta \vDash |\nu|} a_{\nu,\beta}(q)\, C_\beta(1), \qquad a_{\nu,\beta}(q) \in \mathbb{Q}_{\geq 0}[q].$$

By **Theorem A** and $C_j(C_\beta(1)) = C_{(j,\beta)}(1)$,

$$(-1)^{n-\ell(\mu)} m_\mu = \sum_{j,\nu,\beta} b_{\mu;j,\nu}(q)\, a_{\nu,\beta}(q)\, C_{(j,\beta)}(1).$$

Qiu–Zhang (2026a), Theorem 4.2.

Iterating **Theorem A** gives a positive C-expansion

For $\nu \prec \mu$, assume inductively that

$$(-1)^{|\nu|-\ell(\nu)} m_\nu = \sum_{\beta \vDash |\nu|} a_{\nu,\beta}(q)\, C_\beta(1), \qquad a_{\nu,\beta}(q) \in \mathbb{Q}_{\geq 0}[q].$$

By **Theorem A** and $C_j(C_\beta(1)) = C_{(j,\beta)}(1)$,

$$(-1)^{n-\ell(\mu)} m_\mu = \sum_{j,\nu,\beta} b_{\mu;j,\nu}(q)\, a_{\nu,\beta}(q)\, C_{(j,\beta)}(1).$$

Corollary

For every $\mu \vdash n$, we have

$$(-1)^{n-\ell(\mu)} m_\mu = \sum_{\alpha \vDash n} a_{\mu,\alpha}(q)\, C_\alpha(1), \qquad a_{\mu,\alpha}(q) \in \mathbb{Q}_{\geq 0}[q].$$

Qiu–Zhang (2026a), Theorem 4.2.

Nonnegativity: the C-expansion and the shuffle theorem

Positive C-expansion

$$(-1)^{n-\ell(\mu)} m_\mu = \sum_{\alpha \vDash n} a_{\mu,\alpha}(q) C_\alpha(1), \qquad a_{\mu,\alpha}(q) \in \mathbb{Q}_{\geq 0}[q].$$

By the shuffle theorem and LLT positivity, for $r \geq 1$ we have

$$\langle \nabla^r C_\alpha(1), s_\lambda \rangle \in \mathbb{N}[q,t].$$

Carlsson–Mellit, J. Amer. Math. Soc. 31 (2018); Mellit, Duke Math. J. 170 (2021); Grojnowski–Haiman (2007).

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Apply ∇^r

$$(-1)^{n-\ell(\mu)} \langle \nabla^r m_\mu, s_\lambda \rangle = \sum_{\alpha \vDash n} a_{\mu,\alpha}(q) \langle \nabla^r C_\alpha(1), s_\lambda \rangle \in \mathbb{Q}_{\geq 0}[q,t].$$

Carlsson–Mellit, J. Amer. Math. Soc. 31 (2018); Mellit, Duke Math. J. 170 (2021); Grojnowski–Haiman (2007). Qiu–Zhang (2026a), Theorem 1.2.

Integrality and Schur positivity

BHMPS: a $\mathbb{Z}[q,t]$-linear LLT expansion of $\nabla^r s_\rho$. LLT Schur coefficients lie in $\mathbb{Z}[t]$, so

$$\langle \nabla^r s_\rho, s_\lambda \rangle \in \mathbb{Z}[q,t].$$

BHMPS, J. Amer. Math. Soc. 38 (2025), 1049–1106.

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Integral inverse Kostka matrix

$$m_\mu = \sum_{\rho \vdash n} k_{\mu,\rho} s_\rho, \qquad k_{\mu,\rho} \in \mathbb{Z}.$$

$$\langle \nabla^r m_\mu, s_\lambda \rangle = \sum_{\rho \vdash n} k_{\mu,\rho} \langle \nabla^r s_\rho, s_\lambda \rangle \in \mathbb{Z}[q,t].$$

BHMPS, J. Amer. Math. Soc. 38 (2025), 1049–1106. Eğecioğlu–Remmel, Linear Multilinear Algebra 26 (1990), 59–84.

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Nonnegativity + integrality

$$(-1)^{n-\ell(\mu)} \langle \nabla^r m_\mu, s_\lambda \rangle \in \mathbb{Z}[q,t] \cap \mathbb{Q}_{\geq 0}[q,t] = \mathbb{N}[q,t].$$

BHMPS, J. Amer. Math. Soc. 38 (2025), 1049–1106. Eğecioğlu–Remmel, Linear Multilinear Algebra 26 (1990), 59–84.

Example: iterating **Theorem A** for $\mu = (2, 1)$

$$-m_{(2,1)} = 2C_1(-m_{(2)}) + qC_2(m_{(1)}) + (2+q)C_3(1).$$

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$$-m_{(2)} = C_{(1,1)}(1) + (1+q)C_{(2)}(1), \qquad m_{(1)} = C_{(1)}(1).$$

Substitute using $C_j(C_\alpha(1)) = C_{(j,\alpha)}(1)$:

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Nonnegative C-coefficients and the shuffle theorem give Schur positivity of $-\nabla^r m_{(2,1)}$.

Strategy for **Theorem A**: operator identities

Lemma

For $n \geq 1$,

$$(-1)^{n-1} m_{(n)} = \sum_{j=1}^{n-1} q^{j-1} C_j\big((-1)^{n-j-1} m_{(n-j)}\big) + [n]_q C_n(1),$$

where $[n]_q = 1 + q + \cdots + q^{n-1}$.

Qiu–Zhang (2026a), Lemma 2.1. HMZ, Canad. J. Math. 64 (2012), 822–844.

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where $[n]_q = 1 + q + \cdots + q^{n-1}$.

Lemma

For $a, i \geq 1$ and any symmetric function g,

$$(-1)^{a-1} m_{(a)} C_i(g) = C_i\big((-1)^{a-1} m_{(a)} g\big) + (1 - q^a) C_{a+i}(g).$$

The coefficient $1 - q^a$ requires regrouping.

Qiu–Zhang (2026a), Lemma 2.1. Lemma 2.2. HMZ, Canad. J. Math. 64 (2012), 822–844.

Strategy for **Theorem A**: monomial multiplication

Lemma (Multiplication by $m_{(a)}$)

For a positive integer a and a partition μ, we have

$$m_{(a)} m_\mu = \big(1 + \#\{i : \mu_i = a\}\big) m_{\mu \cup \{a\}} + \sum_{j \in \{\mu_1, \ldots, \mu_{\ell(\mu)}\}} \big(1 + \#\{i : \mu_i = a + j\}\big) m_{(\mu \setminus \{j\}) \cup \{a+j\}}.$$

The sum runs over the distinct parts of μ; $\mu \setminus \{j\}$ removes one part j.

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The sum runs over the distinct parts of μ; $\mu\setminus\{j\}$ removes one part j.

$$m_{(2)}m_{(1)} = m_{(2,1)} + m_{(3)}, \qquad -m_{(2,1)} = (-m_{(2)})m_{(1)} + m_{(3)}.$$

Strategy for **Theorem A**: regrouping the terms

Proving the recursion in Theorem A

1. Label repeated parts and choose a smallest part.
2. Apply multiplication, recursion, and commutation.
3. Regroup to obtain nonnegative coefficients.

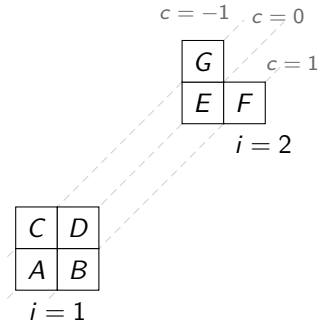
The partitions inside C_j decrease in the order $\prec$.

Divide by multiplicity factorials: $b_{\mu;j,\nu}(q) \in \mathbb{Q}_{\geq 0}[q]$.

Qiu–Zhang (2026a), Theorems 4.1–4.2.

LLT polynomials: attacking pairs

Let $\boldsymbol{\eta} = (\eta^{(1)}, \ldots, \eta^{(k)})$ be a tuple of skew diagrams. The content of a box $u = (x, y)$ is $c(u) = x - y$.



Here $\boldsymbol{\eta} = ((2,2),(2,1))$.

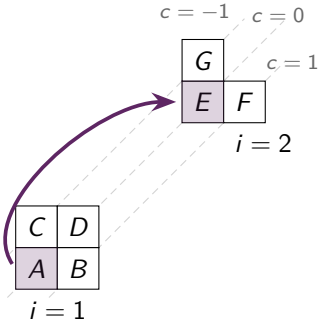
For $u \in \eta^{(i)}$ and $v \in \eta^{(j)}$, the pair (u, v) is *attacking* if

$$\begin{cases} c(v) = c(u), & i < j, \\ c(v) = c(u) + 1, & i > j. \end{cases}$$

Lascoux–Leclerc–Thibon, J. Math. Phys. 38 (1997), 1041–1068.

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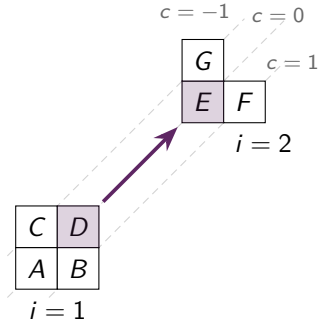
Equal contents: (A, E)

$$c(A) = c(E) = 0, \qquad i = 1 < 2 = j.$$

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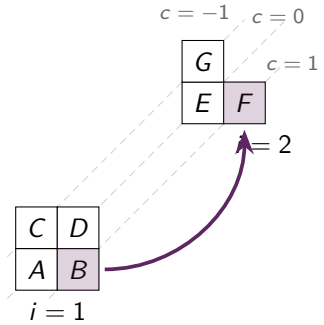
Equal contents: (D, E)

$$c(D) = c(E) = 0, \qquad i = 1 < 2 = j.$$

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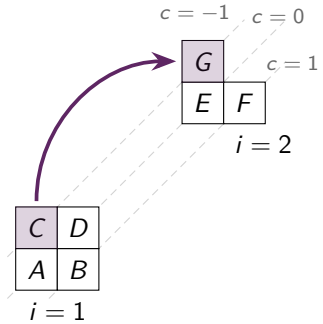
Equal contents: (B, F)

$$c(B) = c(F) = 1, \qquad i = 1 < 2 = j.$$

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LLT polynomials: attacking pairs

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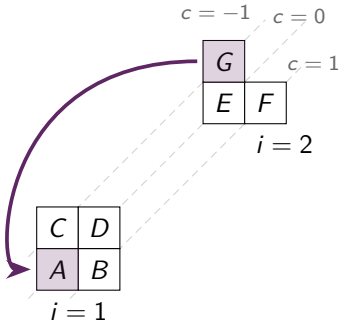
Equal contents: (C, G)

$$c(C) = c(G) = -1, \qquad i = 1 < 2 = j.$$

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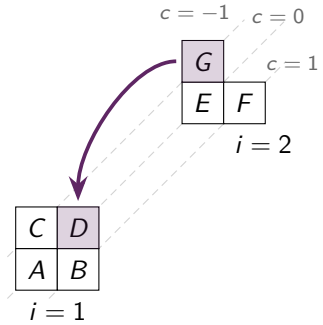
Adjacent contents: (G, A)

$$c(G) = -1, \quad c(A) = 0 = c(G) + 1, \quad i = 2 > 1 = j.$$

Lascoux–Leclerc–Thibon, J. Math. Phys. 38 (1997), 1041–1068.

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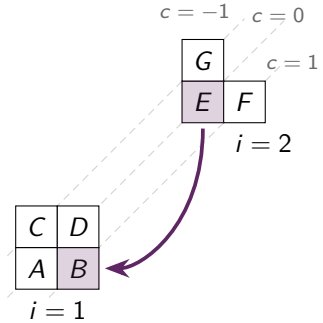
Adjacent contents: (G, D)

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Adjacent contents: (E, B)

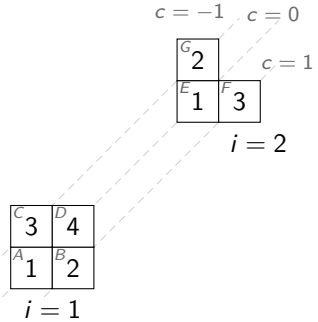
$$c(E) = 0, \quad c(B) = 1 = c(E) + 1, \quad i = 2 > 1 = j.$$

The seven attacking pairs are $(A, E), (D, E), (B, F), (C, G), (G, A), (G, D), (E, B)$.

Lascoux–Leclerc–Thibon, J. Math. Phys. 38 (1997), 1041–1068.

LLT polynomials: inversions in a tableau

Semistandard fillings have positive integer entries, weakly increasing along rows and strictly increasing up columns in each component.



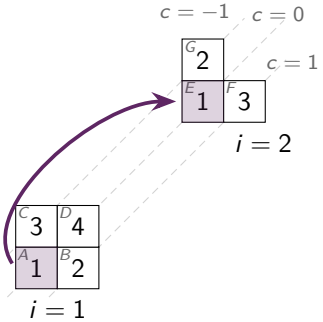
An attacking pair (u, v) is an *inversion* if $T(u) > T(v)$.

$$\mathrm{inv}(T) = \#\{(u, v) \text{ attacking} : T(u) > T(v)\}.$$

Lascoux–Leclerc–Thibon, J. Math. Phys. 38 (1997), 1041–1068.

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Pair (A, E)

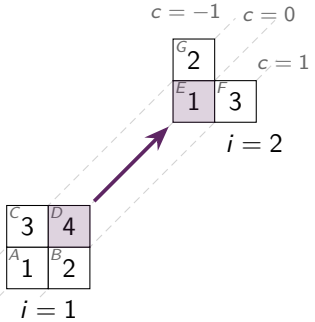
$$T(A) = 1 = 1 = T(E).$$

No inversion; contribution 0. Count so far: 0.

Lascoux–Leclerc–Thibon, J. Math. Phys. 38 (1997), 1041–1068.

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Pair (D, E)

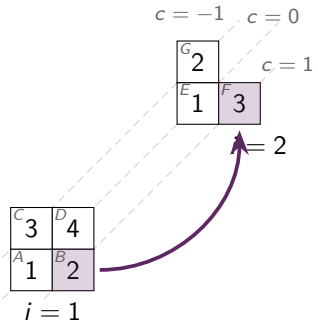
$$T(D) = 4 > 1 = T(E).$$

Inversion; contribution 1. Count so far: 1.

Lascoux–Leclerc–Thibon, J. Math. Phys. 38 (1997), 1041–1068.

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Pair (B, F)

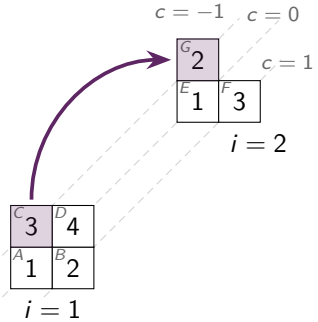
$$T(B) = 2 < 3 = T(F).$$

No inversion; contribution 0. Count so far: 1.

Lascoux–Leclerc–Thibon, J. Math. Phys. 38 (1997), 1041–1068.

LLT polynomials: inversions in a tableau

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Pair (C, G)

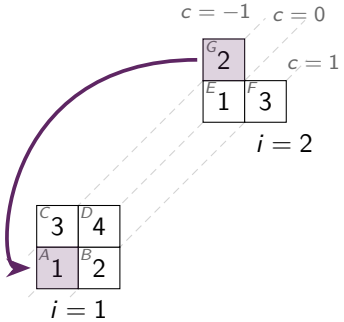
$$T(C) = 3 > 2 = T(G).$$

Inversion; contribution 1. Count so far: 2.

Lascoux–Leclerc–Thibon, J. Math. Phys. 38 (1997), 1041–1068.

LLT polynomials: inversions in a tableau

Semistandard fillings have positive integer entries, weakly increasing along rows and strictly increasing up columns in each component.



An attacking pair (u, v) is an *inversion* if $T(u) > T(v)$.

$$\mathrm{inv}(T) = \#\{(u, v) \text{ attacking} : T(u) > T(v)\}.$$

Pair (G, A)

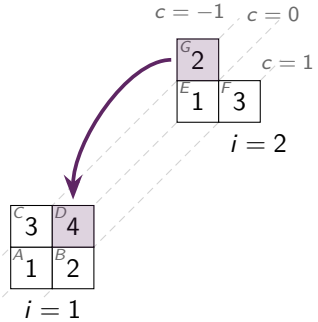
$$T(G) = 2 > 1 = T(A).$$

Inversion; contribution 1. Count so far: 3.

Lascoux–Leclerc–Thibon, J. Math. Phys. 38 (1997), 1041–1068.

LLT polynomials: inversions in a tableau

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An attacking pair (u, v) is an *inversion* if $T(u) > T(v)$.

$$\mathrm{inv}(T) = \#\{(u, v) \text{ attacking} : T(u) > T(v)\}.$$

Pair (G, D)

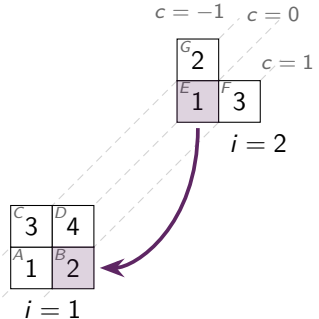
$$T(G) = 2 < 4 = T(D).$$

No inversion; contribution 0. Count so far: 3.

Lascoux–Leclerc–Thibon, J. Math. Phys. 38 (1997), 1041–1068.

LLT polynomials: inversions in a tableau

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Pair (E, B)

$$T(E) = 1 < 2 = T(B).$$

No inversion; contribution 0. Count so far: 3.

$$\mathrm{inv}(T) = 0 + 1 + 0 + 1 + 1 + 0 + 0 = 3.$$

The inversions are $(D, E), (C, G), (G, A)$; the tableau contributes $t^3 x_1^2 x_2^2 x_3^2 x_4$.

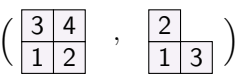
Lascoux–Leclerc–Thibon, J. Math. Phys. 38 (1997), 1041–1068.

The LLT polynomial

Definition (Lascoux–Leclerc–Thibon)

For a tuple of skew diagrams $\boldsymbol{\eta}$, define

$$G_{\boldsymbol{\eta}}[X;t] = \sum_{T \in \mathrm{SSYT}(\boldsymbol{\eta})} t^{\mathrm{inv}(T)} x^T, \qquad x^T = \prod_{u \in \boldsymbol{\eta}} x_{T(u)}.$$

$T =$ 

$\eta^{(1)}$ $\eta^{(2)}$

For the preceding tableau,

$$t^{\mathrm{inv}(T)} x^T = t^3 x_1^2 x_2^2 x_3^2 x_4.$$

At $t = 1$, $G_{\boldsymbol{\eta}}[X;1] = \prod_i s_{\eta^{(i)}}[X]$.

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$$T = \Big(\begin{array}{|c|c|}\hline 3 & 4 \\ \hline 1 & 2 \\ \hline \end{array} \quad , \quad \begin{array}{|c|c|}\hline 2 & \multicolumn{1}{c}{} \\ \hline 1 & 3 \\ \hline \end{array} \Big)$$

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For the preceding tableau,

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At $t = 1$, $G_{\boldsymbol{\eta}}[X;1] = \prod_i s_{\eta^{(i)}}[X]$.

Theorem (Grojnowski–Haiman)

The function $G_{\boldsymbol{\eta}}[X;t]$ is symmetric, and for every partition λ,

$$\langle G_{\boldsymbol{\eta}}[X;t], s_\lambda \rangle \in \mathbb{N}[t].$$

Lascoux–Leclerc–Thibon, J. Math. Phys. 38 (1997), 1041–1068. Grojnowski–Haiman, unpublished manuscript (2007).

The skew tuple associated with a cover

Let $\mu \succcurlyeq \nu$ in degree n, with $\nu = \mu - \epsilon_i + \epsilon_j$ and $i < j$. Pad both partitions with zeros to length $R \geq n$. Define

$$\theta^{\mu,\nu} = \left(\frac{\nu_1 - 1}{-1}, \ldots, \frac{\nu_{i-1} - 1}{-1}, \frac{(\nu_i, \nu_{i+1})}{(0,0)}, \frac{\nu_{i+2}}{0}, \ldots, \frac{\nu_R}{0} \right).$$

The superscript R means transpose and rotate each component by 180°, preserving contents and component order.

Gu–Kang–Qiu–Zhang (2026), Section 4.2 and Theorem 4.14.

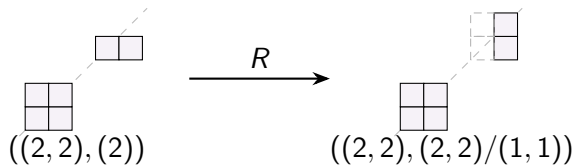
The skew tuple associated with a cover

Let $\mu \gtrdot \nu$ in degree n, with $\nu = \mu - \epsilon_i + \epsilon_j$ and $i < j$. Pad both partitions with zeros to length $R \geq n$. Define

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The superscript R means transpose and rotate each component by 180°, preserving contents and component order.

For $\mu = (3,2,1)$ and $\nu = (2,2,2)$, the transformation is



Gu–Kang–Qiu–Zhang (2026), Section 4.2 and Theorem 4.14.

An LLT formula for a covering difference

Set $\theta = \theta^{\mu,\nu}$, $\sigma_0 = w_0 \in \mathfrak{S}_{R-1}$, $\sigma_0(\beta/\alpha) = \theta$, and $h = h_{\sigma_0}(\beta/\alpha)$.

Theorem B (Gu–Kang–Qiu–Zhang)

If $\mu \gtrdot \nu$, then

$$\omega \widetilde{H}_\mu[X; 0, t^{-1}] - \omega \widetilde{H}_\nu[X; 0, t^{-1}] = -t^{h-A(\theta)-n(\nu)} G_{\theta^R}[X; t].$$

Gu–Kang–Qiu–Zhang (2026), Theorem 4.14.

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- $n(\nu) = \sum_a (a-1)\nu_a$: Hall–Littlewood normalization.
- $A(\theta)$: attack count; $\omega G_\theta[X; t^{-1}] = t^{-A(\theta)} G_{\theta^R}[X; t]$.
- h: the triple count in LLT symmetrization.

Gu–Kang–Qiu–Zhang (2026), Theorem 4.14.

Weyl symmetrization

For $z^\gamma = z_1^{\gamma_1} \cdots z_R^{\gamma_R}$,

$$\boldsymbol{\sigma}_R(f) = \sum_{w \in \mathfrak{S}_R} w\left(\frac{f}{\prod_{a<b}(1 - z_b/z_a)}\right).$$

For a partition λ,

$$\boldsymbol{\sigma}_R(z^\lambda) = s_\lambda(z_1, \ldots, z_R).$$

For example, $\boldsymbol{\sigma}_2(z_1^2) = z_1^2 + z_1 z_2 + z_2^2$.

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For example, $\boldsymbol{\sigma}_2(z_1^2) = z_1^2 + z_1 z_2 + z_2^2$.

Polynomial part

Retain irreducible GL_R characters with partition highest weights and identify them with $s_\lambda[X]$.

$$\left(\boldsymbol{\sigma}_R(z^\gamma)\right)_{\mathrm{pol}} = s_\gamma[X] = \det\left(h_{\gamma_a - a + b}[X]\right)_{1 \le a,b \le R}.$$

For $\gamma \in \mathbb{Z}^R$, s_γ straightens to a signed Schur function or zero.

LLT symmetrization for a covering pair

For $\mu \gg \nu$, pad ν to length $R \geq n$ and set

$$\Omega_i = \{(a,b) : 1 \leq a < b \leq R\} \setminus \{(i, i+1)\}.$$

Proposition (BHMPS)

For $\theta = \theta^{\mu,\nu}$, we have

$$\left(\sigma_R \left(\frac{z^\nu}{\prod_{(a,b)\in\Omega_i}(1 - t z_a / z_b)} \right) \right)_{\mathrm{pol}} = t^h G_\theta[X; t^{-1}].$$

BHMPS, J. Amer. Math. Soc. 38 (2025), Proposition 5.4.3 and Theorem 5.5.4; specialized to $\theta^{\mu,\nu}$.

Hall–Littlewood symmetrization

Proposition (Macdonald)

For $\lambda \vdash n$ and $R \geq n$, we have

$$t^{n(\lambda)} \widetilde{H}_\lambda[X; 0, t^{-1}] = \left(\sigma_R \left(\frac{z^\lambda}{\prod_{a<b}(1 - tz_a/z_b)} \right) \right)_{\mathrm{pol}}.$$

Geometric expansion:

$$\frac{1}{1 - tz_a/z_b} = \sum_{d \geq 0} t^d z_a^d z_b^{-d}.$$

Macdonald, Symmetric Functions and Hall Polynomials, 2nd ed., Oxford, 1995, Chapter III, (6.3).

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Geometric expansion:

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Schur straightening gives

$$t^{n(\lambda)} \widetilde{H}_\lambda[X; 0, t^{-1}] = \sum_{(d_{ab}) \geq 0} t^{\sum_{a<b} d_{ab}} s_{\lambda + \sum_{a<b} d_{ab}(\epsilon_a - \epsilon_b)}[X].$$

Macdonald, Symmetric Functions and Hall Polynomials, 2nd ed., Oxford, 1995, Chapter III, (6.3).

Strategy for **Theorem B**: cancellation and ω

For $\mu \succcurlyeq \nu = \mu - \epsilon_i + \epsilon_j$, we have

$$t^{n(\nu)}\big(\widetilde{H}_\nu[X;0,t^{-1}] - \widetilde{H}_\mu[X;0,t^{-1}]\big) = \left(\boldsymbol{\sigma}_R\left(\frac{z^\nu\,(1-tz_i/z_{i+1})}{\prod_{a<b}(1-tz_a/z_b)}\right)\right)_{\mathrm{pol}}$$
$$= \left(\boldsymbol{\sigma}_R\left(\frac{z^\nu}{\prod_{(a,b)\in\Omega_i}(1-tz_a/z_b)}\right)\right)_{\mathrm{pol}}.$$

Gu–Kang–Qiu–Zhang (2026), Sections 4.1–4.2.

Strategy for **Theorem B**: cancellation and ω

For $\mu \succcurlyeq \nu = \mu - \epsilon_i + \epsilon_j$, we have

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$$= \left(\boldsymbol{\sigma}_R\left(\frac{z^\nu}{\prod_{(a,b)\in\Omega_i}(1-tz_a/z_b)}\right)\right)_{\mathrm{pol}}.$$

Recall the formula on p. 33 (BHMPS):

$$\left(\boldsymbol{\sigma}_R\left(\frac{z^\nu}{\prod_{(a,b)\in\Omega_i}(1-tz_a/z_b)}\right)\right)_{\mathrm{pol}} = t^h G_\theta[X;t^{-1}].$$

Gu–Kang–Qiu–Zhang (2026), Sections 4.1–4.2. BHMPS, J. Amer. Math. Soc. 38 (2025), Proposition 5.4.3 and Theorem 5.5.4.

Strategy for **Theorem B**: cancellation and ω

For $\mu \gg \nu = \mu - \epsilon_i + \epsilon_j$, we have

$$t^{n(\nu)}\big(\widetilde{H}_\nu[X;0,t^{-1}] - \widetilde{H}_\mu[X;0,t^{-1}]\big) = \left(\boldsymbol{\sigma}_R\left(\frac{z^\nu\,(1-tz_i/z_{i+1})}{\prod_{a<b}(1-tz_a/z_b)}\right)\right)_{\mathrm{pol}}$$
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Using $\omega G_\theta[X;t^{-1}] = t^{-A(\theta)}G_{\theta^R}[X;t]$, we obtain

Theorem B (Gu–Kang–Qiu–Zhang)

$$\omega\widetilde{H}_\mu[X;0,t^{-1}] - \omega\widetilde{H}_\nu[X;0,t^{-1}] = -t^{h-A(\theta)-n(\nu)}G_{\theta^R}[X;t].$$

Gu–Kang–Qiu–Zhang (2026), Sections 4.1–4.2. BHMPS, J. Amer. Math. Soc. 38 (2025), Proposition 5.4.3 and Theorem 5.5.4.

The Catalanimal formula for ∇^r

For $\mu \gtrdot \nu = \mu - \epsilon_i + \epsilon_j$, use $G_{\boldsymbol{\nu}^{\mu,\nu}} = G_{(\theta^{\mu,\nu})^R}$ and its constant-offset r-stretch:

$$H^{r,1}_{\boldsymbol{\nu}_r}(z;q,t) = \boldsymbol{\sigma}_{rn}\left(z^{\boldsymbol{\kappa}^{\mathrm{cat}}}\frac{\prod_{\alpha\in R_{qt}}(1-qtz^{\alpha})}{\prod_{\alpha\in R_q}(1-qz^{\alpha})\prod_{\alpha\in R_t}(1-tz^{\alpha})}\right).$$

BHMPS, J. Reine Angew. Math. 811 (2024), 93–133.

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Theorem (BHMPS, specialized to a covering pair)

For $r \geq 1$,

$$\nabla^r G_{\boldsymbol{\nu}^{\mu,\nu}}[X;t] = -(qt)^{1+r(\nu_{i+1}-1)} t^{A_r(\boldsymbol{\nu}^{\mu,\nu})} \omega\Big(\big(H^{r,1}_{\boldsymbol{\nu}_r}(z;t,q)\big)_{\mathrm{pol}} \Big).$$

A_r: before stretching, equal-content attacks have weight r; all others have weight 1. Evaluate the Catalanimal at (t,q).

BHMPS, J. Reine Angew. Math. 811 (2024), 93–133. Gu–Kang–Qiu–Zhang (2026), Theorem 5.1.

BHMPS: ∇^r and LLT expansions

$$\nabla^r G_{\boldsymbol{\nu}^{\mu,\nu}}[X;t] \xrightarrow{\text{Catalanimal formula}} \omega\left(\left(H^{r,1}_{\nu_r}(z;t,q)\right)_{\mathrm{pol}}\right)$$

BHMPS, J. Reine Angew. Math. 811 (2024), 93–133.

BHMPS: ∇^r and LLT expansions

$$\nabla^r G_{\boldsymbol{\nu}^{\mu,\nu}}[X;t] \xrightarrow{\text{Catalanimal formula}} \omega\left(\left(H^{r,1}_{\nu_r}(z;t,q)\right)_{\mathrm{pol}}\right)$$

$$\xrightarrow{\text{Cauchy identity and the action of } \omega} \text{an LLT expansion.}$$

BHMPS, J. Reine Angew. Math. 811 (2024), 93–133. J. Amer. Math. Soc. 38 (2025), 1049–1106.

The LLT expansion after applying ∇^r

B: boundary between component diagonals; a_B, b_B: target and source sizes. Cauchy data: $\alpha(\boldsymbol{\lambda}), \beta(\boldsymbol{\lambda}), \theta_r$; $|\boldsymbol{\lambda}| = \sum_B |\lambda_B|$.
$e^{\mu,\nu} = h_{\sigma_0}(\beta/\alpha) - A(\theta^{\mu,\nu}) - n(\nu)$ and

$$\varphi(\boldsymbol{\lambda}) = h_{\theta_r}(\beta(\boldsymbol{\lambda})/\alpha(\boldsymbol{\lambda})) - A(\theta_r(\beta(\boldsymbol{\lambda})/\alpha(\boldsymbol{\lambda}))) .$$

Theorem C (Gu–Kang–Qiu–Zhang)

For $\mu \gg \nu$ and $r \geq 1$,

$$\nabla^r \left(\omega \widetilde{H}_\mu[X; 0, t^{-1}] - \omega \widetilde{H}_\nu[X; 0, t^{-1}] \right)$$
$$= (qt)^{1+r(\nu_{i+1}-1)} t^{e^{\mu,\nu}+A_r(\boldsymbol{\nu}^{\mu,\nu})} \sum_{\substack{\boldsymbol{\lambda}=(\lambda_B) \\ \ell(\lambda_B)\leq \min(a_B,b_B) \\ \alpha(\boldsymbol{\lambda})\leq\beta(\boldsymbol{\lambda})}} q^{|\boldsymbol{\lambda}|} t^{\varphi(\boldsymbol{\lambda})} G_{(\theta_r(\beta(\boldsymbol{\lambda})/\alpha(\boldsymbol{\lambda})))^R}[X; t].$$

Gu–Kang–Qiu–Zhang (2026), Theorem 5.11 and Section 5.2.

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For $\mu \gg \nu$ and $r \geq 1$,

$$\begin{aligned} &\nabla^r \left(\omega \widetilde{H}_\mu[X; 0, t^{-1}] - \omega \widetilde{H}_\nu[X; 0, t^{-1}] \right) \\ &= (qt)^{1+r(\nu_{i+1}-1)} t^{e^{\mu,\nu}+A_r(\boldsymbol{\nu}^{\mu,\nu})} \sum_{\substack{\boldsymbol{\lambda}=(\lambda_B) \\ \ell(\lambda_B)\leq \min(a_B,b_B) \\ \alpha(\boldsymbol{\lambda})\leq\beta(\boldsymbol{\lambda})}} q^{|\boldsymbol{\lambda}|} t^{\varphi(\boldsymbol{\lambda})} G_{(\theta_r(\beta(\boldsymbol{\lambda})/\alpha(\boldsymbol{\lambda})))^R}[X; t]. \end{aligned}$$

Finite sum;

Gu–Kang–Qiu–Zhang (2026), Theorem 5.11 and Section 5.2.

The LLT expansion after applying ∇^r

B: boundary between component diagonals; a_B, b_B: target and source sizes. Cauchy data: $\alpha(\boldsymbol{\lambda}), \beta(\boldsymbol{\lambda}), \theta_r$; $|\boldsymbol{\lambda}| = \sum_B |\lambda_B|$.
$e^{\mu,\nu} = h_{\sigma_0}(\beta/\alpha) - A(\theta^{\mu,\nu}) - n(\nu)$ and

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For $\mu \gg \nu$ and $r \geq 1$,

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Finite sum; Schur coefficients in $\mathbb{N}[q, t^{\pm 1}]$.

Gu–Kang–Qiu–Zhang (2026), Theorem 5.11 and Section 5.2.

Example: the four-term LLT expansion

$$
\begin{aligned}
&\nabla\Big(\omega\widetilde{H}_{(3,2,1)}[X;0,t^{-1}]-\omega\widetilde{H}_{(2,2,2)}[X;0,t^{-1}]\Big)\\
&=q^2t^{-2}G_{(\varnothing,(2),(1),(2,2)/(1,1),(1))}[X;t]\\
&+q^3t^{-3}G_{(\varnothing,(2),\varnothing,(2,2)/(1,1),(2,2)/(1,1))}[X;t]\\
&+q^3t^{-1}G_{(\varnothing,(1),(1),(3,3,3)/(2,2,2),(1))}[X;t]\\
&+q^4t^{-2}G_{(\varnothing,(1),\varnothing,(3,3,3)/(2,2,2),(2,2)/(1,1))}[X;t].
\end{aligned}
$$

Gu–Kang–Qiu–Zhang (2026), equation (42).

The HMZ–Carlsson–Mellit formula

Let $\mu \vdash n$, $m = \ell(\mu)$, and $\alpha = (\mu_m, \dots, \mu_1)$. For $\beta = (\beta^{(1)}|\cdots|\beta^{(m)}) \preceq \alpha$, with $\beta^{(j)} \vDash \alpha_j$, set

$$\delta_\mu(\beta) = \textstyle\sum_{j=1}^{m} (m-j)\big(\alpha_j - \ell(\beta^{(j)})\big).$$

Theorem (Haglund–Morse–Zabrocki)

$$\omega \widetilde{H}_\mu[X; 0, t^{-1}] = \sum_{\beta \preceq \alpha} t^{-\delta_\mu(\beta)} C_\beta[X; t].$$

HMZ, Canad. J. Math. 64 (2012), Theorem 5.6.

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Theorem (Haglund–Morse–Zabrocki; Carlsson–Mellit)

$$\nabla \omega \widetilde{H}_\mu[X; 0, t^{-1}] = \sum_{\beta \preceq \alpha} \sum_{\substack{\pi \in \mathrm{LD}_{n,n} \\ \mathrm{comp}(\pi) = \beta}} q^{\mathrm{area}(\pi)} t^{\mathrm{dinv}(\pi) - \delta_\mu(\beta)} F_{\mathrm{ides}(\pi)}[X].$$

Here $\mathrm{LD}_{n,n}$ *denotes standard labeled Dyck paths;* comp *records diagonal returns, and* F_S *is a fundamental quasisymmetric function.*

HMZ, Canad. J. Math. 64 (2012), Theorem 5.6. Carlsson–Mellit, J. Amer. Math. Soc. 31 (2018); GKQZ (2026), Theorem 3.6.

Polynomiality of $\nabla\omega\widetilde{H}_\mu[X;0,t^{-1}]$

$$\nabla\omega\widetilde{H}_\mu[X;0,t^{-1}] = \sum_{\beta\preceq\alpha} \sum_{\substack{\pi\in \mathrm{LD}_{n,n} \\ \mathrm{comp}(\pi)=\beta}} q^{\mathrm{area}(\pi)} t^{\mathrm{dinv}(\pi)-\delta_\mu(\beta)} F_{\mathrm{ides}(\pi)}[X].$$

Lemma (Gu–Kang–Qiu–Zhang)

For $\pi \in \mathrm{LD}_{n,n}$ with $\mathrm{comp}(\pi) = \beta \preceq \alpha$, we have

$$\mathrm{dinv}(\pi) \geq \delta_\mu(\beta).$$

HMZ, Canad. J. Math. 64 (2012); Carlsson–Mellit, J. Amer. Math. Soc. 31 (2018); GKQZ (2026), Lemma 3.7.

Polynomiality of $\nabla\omega\widetilde{H}_\mu[X;0,t^{-1}]$

$$\nabla\omega\widetilde{H}_\mu[X;0,t^{-1}] = \sum_{\beta\preceq\alpha} \sum_{\substack{\pi\in \mathrm{LD}_{n,n} \\ \mathrm{comp}(\pi)=\beta}} q^{\mathrm{area}(\pi)} t^{\mathrm{dinv}(\pi)-\delta_\mu(\beta)} F_{\mathrm{ides}(\pi)}[X].$$

Lemma (Gu–Kang–Qiu–Zhang)

For $\pi \in \mathrm{LD}_{n,n}$ with $\mathrm{comp}(\pi) = \beta \preceq \alpha$, we have

$$\mathrm{dinv}(\pi) \geq \delta_\mu(\beta).$$

Hence $[m_\lambda]\,\nabla\omega\widetilde{H}_\mu[X;0,t^{-1}] \in \mathbb{N}[q,t]$. The inverse Kostka matrix has integer entries, so

Corollary (Gu–Kang–Qiu–Zhang)

For $\mu, \lambda \vdash n$, we have

$$\left\langle \nabla\omega\widetilde{H}_\mu[X;0,t^{-1}], s_\lambda \right\rangle \in \mathbb{Z}[q,t].$$

HMZ, Canad. J. Math. 64 (2012); Carlsson–Mellit, J. Amer. Math. Soc. 31 (2018); GKQZ (2026), Lemma 3.7. GKQZ (2026), Corollary 3.8; Eğecioğlu–Remmel, Linear Multilinear Algebra 26 (1990), 59–84.

Schur positivity for Hall–Littlewood covering differences

Let $\mu \gg \nu$ in degree n and $r \in \mathbb{Z}_{\geq 1}$. BHMPS give $\langle \nabla s_\rho, s_\lambda \rangle \in \mathbb{Z}[q,t]$. Applying ∇^{r-1} to the preceding corollary gives

$$\left\langle \nabla^r\big(\omega \widetilde{H}_\mu[X;0,t^{-1}] - \omega \widetilde{H}_\nu[X;0,t^{-1}]\big), s_\lambda \right\rangle \in \mathbb{Z}[q,t].$$

BHMPS, J. Amer. Math. Soc. 38 (2025), 1049–1106.

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By **Theorem C** and LLT Schur positivity,

$$\left\langle \nabla^r\big(\omega \widetilde{H}_\mu[X;0,t^{-1}] - \omega \widetilde{H}_\nu[X;0,t^{-1}]\big), s_\lambda \right\rangle \in \mathbb{N}[q,t^{\pm 1}].$$

BHMPS, J. Amer. Math. Soc. 38 (2025), 1049–1106. Grojnowski–Haiman (2007); GKQZ (2026), Theorem 5.11.

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Since $\mathbb{Z}[q,t] \cap \mathbb{N}[q,t^{\pm 1}] = \mathbb{N}[q,t]$, we obtain

Theorem (Gu–Kang–Qiu–Zhang)

For every $\lambda \vdash n$, we have

$$\left\langle \nabla^r\big(\omega \widetilde{H}_\mu[X;0,t^{-1}] - \omega \widetilde{H}_\nu[X;0,t^{-1}]\big), s_\lambda \right\rangle \in \mathbb{N}[q,t].$$

BHMPS, J. Amer. Math. Soc. 38 (2025), 1049–1106. Grojnowski–Haiman (2007); GKQZ (2026), Theorem 5.11. GKQZ (2026), Corollary 5.12.

From covers to arbitrary dominance pairs

Let $\mu, \nu \vdash n$ with $\mu \trianglerighteq \nu$, and let $r \geq 1$. Choose a saturated chain:

$$\mu = \mu^{(0)} \trianglerighteq \mu^{(1)} \trianglerighteq \cdots \trianglerighteq \mu^{(s)} = \nu.$$

Telescoping:

$$\begin{aligned} &\nabla^r \left(\omega \widetilde{H}_\mu[X; 0, t^{-1}] - \omega \widetilde{H}_\nu[X; 0, t^{-1}] \right) \\ &= \sum_{j=0}^{s-1} \nabla^r \left(\omega \widetilde{H}_{\mu^{(j)}}[X; 0, t^{-1}] - \omega \widetilde{H}_{\mu^{(j+1)}}[X; 0, t^{-1}] \right). \end{aligned}$$

Gu–Kang–Qiu–Zhang (2026), Corollary 5.12.

From covers to arbitrary dominance pairs

Let $\mu, \nu \vdash n$ with $\mu \trianglerighteq \nu$, and let $r \geq 1$. Choose a saturated chain:

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Each summand is Schur positive. Thus, for every $\lambda \vdash n$,

$$\left\langle \nabla^r \left(\omega \widetilde{H}_\mu[X; 0, t^{-1}] - \omega \widetilde{H}_\nu[X; 0, t^{-1}] \right), s_\lambda \right\rangle \in \mathbb{N}[q, t].$$

Gu–Kang–Qiu–Zhang (2026), Corollary 5.12.

Sergel's square paths theorem

$p_n = m_{(n)} = \sum_{i \geq 1} x_i^n$.

Theorem (Sergel)

$$(-1)^{n-1} \nabla p_n = \sum_{\pi \in \mathrm{Pr}_{n,n}} q^{\mathrm{dinv}(\pi)} t^{\mathrm{area}(\pi)} x_\pi \qquad (n \geq 1).$$

$\mathrm{Pr}_{n,n}$: paths $(0,0) \to (n,n)$ ending East, with positive integer labels strictly increasing up each column.

Loehr–Warrington, Trans. Amer. Math. Soc. 359 (2007), 649–669; Sergel, J. Combin. Theory Ser. A 152 (2017), 363–379.

Sergel's square paths theorem

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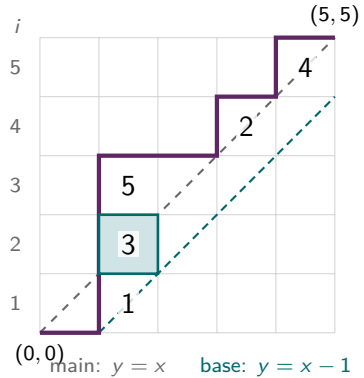
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$\mathrm{Pr}_{n,n}$: paths $(0,0) \to (n,n)$ ending East, with positive integer labels strictly increasing up each column. Paths may pass below $y = x$.

Loehr–Warrington, Trans. Amer. Math. Soc. 359 (2007), 649–669; Sergel, J. Combin. Theory Ser. A 152 (2017), 363–379.

Square paths: area



Indices i at left; labels w_i in the cells.

North step i starts at $(u_i, i-1)$: $a_i = i - 1 - u_i$.

$$(a_1, \ldots, a_5) = (-1, 0, 1, 0, 0).$$

$s = -\min_i a_i = 1$; base diagonal $y = x - 1$.

$$\text{area}(\pi) = \sum_{i=1}^{5} (a_i + s).$$

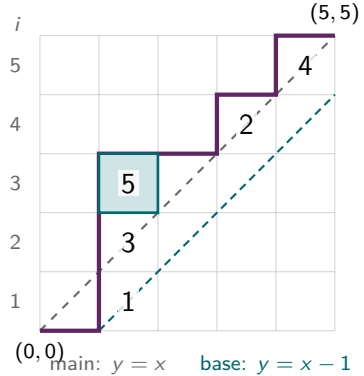
Row contributions: $(0, 1, 2, 1, 1)$.

Cell $(1, 1)$
Row 2: one cell.

Cell coordinates: lower-left corners.

Sergel, J. Combin. Theory Ser. A 152 (2017), 363–379; Qiu–Zhang (2026b), Section 2.2.

Square paths: area



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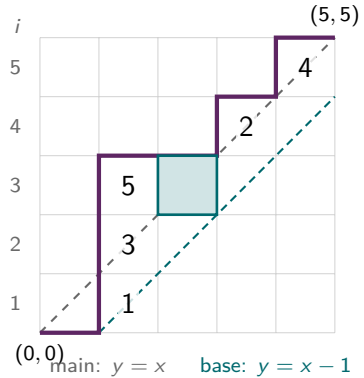
Row contributions: $(0, 1, 2, 1, 1)$.

Cell $(1, 2)$
Row 3: first cell.

Cell coordinates: lower-left corners.

Sergel, J. Combin. Theory Ser. A 152 (2017), 363–379; Qiu–Zhang (2026b), Section 2.2.

Square paths: area



Indices i at left; labels w_i in the cells.

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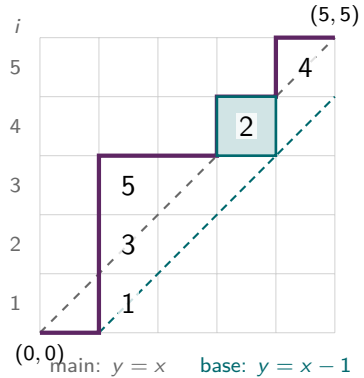
Row contributions: $(0, 1, 2, 1, 1)$.

Cell $(2, 2)$
Row 3: second cell.

Cell coordinates: lower-left corners.

Sergel, J. Combin. Theory Ser. A 152 (2017), 363–379; Qiu–Zhang (2026b), Section 2.2.

Square paths: area



Indices i at left; labels w_i in the cells.

North step i starts at $(u_i, i-1)$: $a_i = i - 1 - u_i$.

$$(a_1, \ldots, a_5) = (-1, 0, 1, 0, 0).$$

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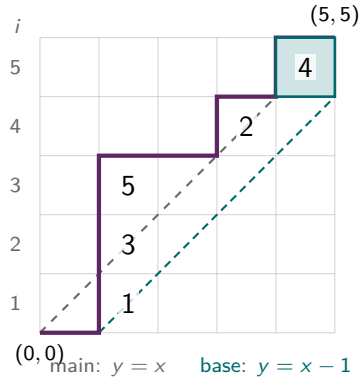
Row contributions: $(0, 1, 2, 1, 1)$.

Cell $(3, 3)$
Row 4: one cell.

Cell coordinates: lower-left corners.

Sergel, J. Combin. Theory Ser. A 152 (2017), 363–379; Qiu–Zhang (2026b), Section 2.2.

Square paths: area



Indices i at left; labels w_i in the cells.

North step i starts at $(u_i, i-1)$: $a_i = i - 1 - u_i$.

$$(a_1, \ldots, a_5) = (-1, 0, 1, 0, 0).$$

$s = -\min_i a_i = 1$; base diagonal $y = x - 1$.

$$\text{area}(\pi) = \sum_{i=1}^{5} (a_i + s).$$

Row contributions: $(0, 1, 2, 1, 1)$.

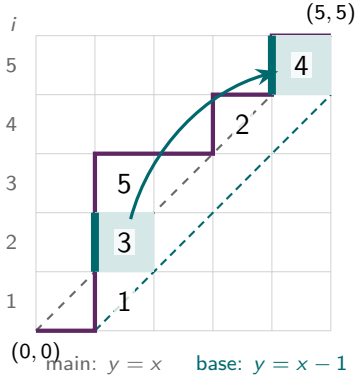
Cell $(4, 4)$
Row 5: one cell.

$$\text{area}(\pi) = 0 + 1 + 2 + 1 + 1 = 5.$$

Cell coordinates: lower-left corners.

Sergel, J. Combin. Theory Ser. A 152 (2017), 363–379; Qiu–Zhang (2026b), Section 2.2.

Primary diagonal inversions



Indices i at left; labels w_i in the cells.

Primary inversions: same diagonal

$$i < j, \qquad a_i = a_j, \qquad w_i < w_j.$$

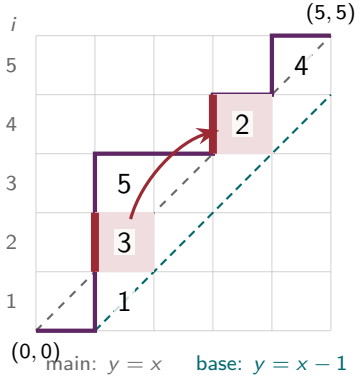
Pair $(2, 5)$

$$a_2 = a_5 = 0, \qquad w_2 = 3 < 4 = w_5.$$

Counted.

Sergel, J. Combin. Theory Ser. A 152 (2017), 363–379; Qiu–Zhang (2026b), Section 2.2.

Primary diagonal inversions



Indices i at left; labels w_i in the cells.

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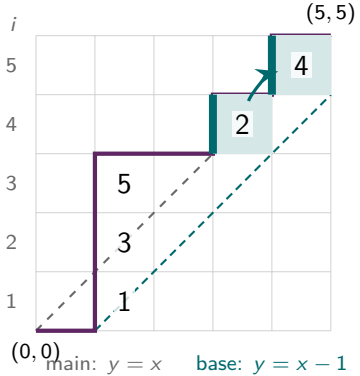
Pair $(2, 4)$

$$a_2 = a_4 = 0, \qquad w_2 = 3 > 2 = w_4.$$

Not counted: labels decrease.

Sergel, J. Combin. Theory Ser. A 152 (2017), 363–379; Qiu–Zhang (2026b), Section 2.2.

Primary diagonal inversions



Indices i at left; labels w_i in the cells.

Primary inversions: same diagonal

$$i < j, \qquad a_i = a_j, \qquad w_i < w_j.$$

Pair $(4, 5)$

$$a_4 = a_5 = 0, \qquad w_4 = 2 < 4 = w_5.$$

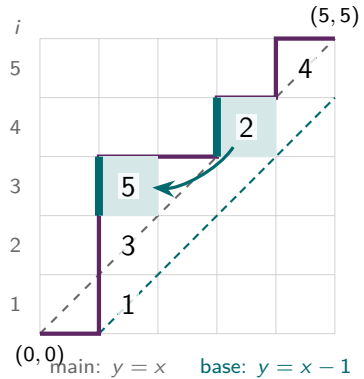
Counted.

Primary pairs:

$$\{(2, 5), (4, 5)\}, \qquad \text{primary count} = 2.$$

Sergel, J. Combin. Theory Ser. A 152 (2017), 363–379; Qiu–Zhang (2026b), Section 2.2.

Secondary diagonal inversions



Indices i at left; labels w_i in the cells.

Secondary inversions: adjacent diagonals

$$i > j, \qquad a_j = a_i + 1, \qquad w_i < w_j.$$

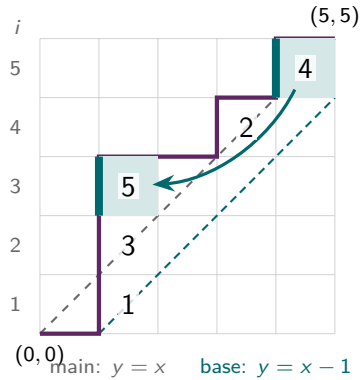
Arrow: later step $\longrightarrow$ earlier step.
Pair $(4, 3)$

$$a_3 = 1 = a_4 + 1, \qquad w_4 = 2 < 5 = w_3.$$

Counted.

Sergel, J. Combin. Theory Ser. A 152 (2017), 363–379; Qiu–Zhang (2026b), Section 2.2.

Secondary diagonal inversions



Indices i at left; labels w_i in the cells.

Secondary inversions: adjacent diagonals

$$i > j, \qquad a_j = a_i + 1, \qquad w_i < w_j.$$

Arrow: later step $\longrightarrow$ earlier step.
Pair $(5, 3)$

$$a_3 = 1 = a_5 + 1, \qquad w_5 = 4 < 5 = w_3.$$

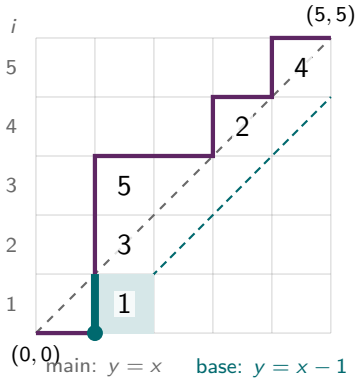
Counted.

Secondary pairs:

$$\{(4, 3), (5, 3)\}, \qquad \text{secondary count} = 2.$$

Sergel, J. Combin. Theory Ser. A 152 (2017), 363–379; Qiu–Zhang (2026b), Section 2.2.

Bonus inversions



Indices i at left; labels w_i in the cells.

Bonus: lower endpoint below $y = x$

$$a_i < 0.$$

North step $i = 1$

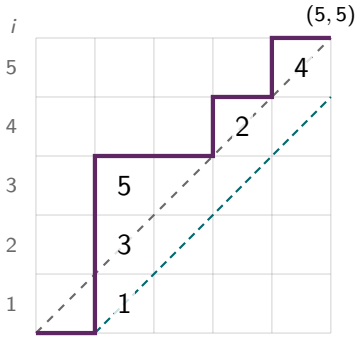
$$(u_1, 0) = (1, 0), \qquad a_1 = -1 < 0.$$

One bonus inversion.

$a_2, a_3, a_4, a_5 \geq 0$.

Sergel, J. Combin. Theory Ser. A 152 (2017), 363–379; Qiu–Zhang (2026b), Section 2.2.

Bonus inversions



Indices i at left; labels w_i in the cells.

Bonus: lower endpoint below $y = x$

$$a_i < 0.$$

Bonus steps: $\{1\}$; bonus count: 1.

$$\mathrm{dinv}(\pi) = \underbrace{2}_{\text{primary}} + \underbrace{2}_{\text{secondary}} + \underbrace{1}_{\text{bonus}} = 5.$$

$$(\mathrm{area}(\pi), \mathrm{dinv}(\pi)) = (5, 5).$$

Sergel, J. Combin. Theory Ser. A 152 (2017), 363–379; Qiu–Zhang (2026b), Section 2.2.

The (km, kn)-square paths theorem

Let $k, m, n \geq 1$ with $\gcd(m, n) = 1$, and set $M = (1 - q)(1 - t)$. Here $f[-MX^{m,n}]$ is the elliptic Hall algebra operator in direction (m, n).

Theorem D (Qiu–Zhang)

$$(-1)^{k-1} \frac{[km]_q}{[k]_q} \, p_k[-MX^{m,n}] \cdot 1 = \sum_{\pi \in \mathrm{Pr}_{km,kn}} q^{\mathrm{dinv}(\pi)} t^{\mathrm{area}(\pi)} x_\pi .$$

$[a]_q = 1 + q + \cdots + q^{a-1}$; $[km]_q / [k]_q = [m]_{q^k}$.

Qiu–Zhang (2026b), Theorems 1.2 and 4.3.

The (km, kn)-square paths theorem

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$[a]_q = 1 + q + \cdots + q^{a-1}$; $[km]_q / [k]_q = [m]_{q^k}$.

Normalization

$$f[-MX^{0,1}] \cdot 1 = f[X], \qquad \nabla f[-MX^{m,n}] \nabla^{-1} = f[-MX^{m+n,n}].$$

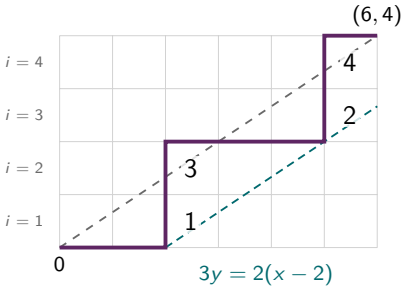
The direction $(m, 1)$

$$f[-MX^{m,1}] \cdot 1 = \nabla^m f, \qquad p_k[-MX^{1,1}] \cdot 1 = \nabla p_k .$$

For $m = n = 1$, this is Sergel's theorem.

Qiu–Zhang (2026b), Theorems 1.2 and 4.3. BHMPS, J. Reine Angew. Math. 811 (2024), 93–133.

(km, kn)-preference functions: area and touches



A path in $\mathrm{Pr}_{km,kn}$ goes from $(0,0)$ to (km, kn) and ends East. Positive integer labels strictly increase up vertical runs.

If North step i starts at $(u_i, i-1)$, set

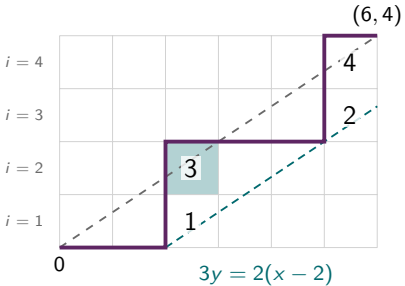
$$a_i = \frac{m}{n}(i-1) - u_i, \qquad s = -\min_i a_i.$$

The base diagonal is $my = n(x-s)$.

$$\mathrm{area}(\pi) = \sum_i \lfloor a_i + s \rfloor, \quad \mathrm{touch}(\pi) = \#\{i : a_i + s = 0\}.$$

Iraci–Pagaria–Paolini–Vanden Wyngaerd, Combin. Theory 3 (2023), no. 2, Paper 4; Qiu–Zhang (2026b), Section 2.2.

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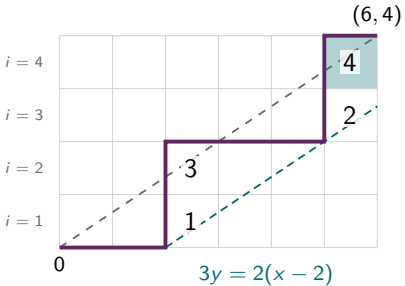
$$\mathrm{area}(\pi) = \sum_i \lfloor a_i + s \rfloor, \quad \mathrm{touch}(\pi) = \#\{i : a_i + s = 0\}.$$

Area: row 2

$$a = (-2, -\tfrac{1}{2}, -2, -\tfrac{1}{2}), \quad s = 2, \qquad \lfloor a_2 + s \rfloor = \lfloor \tfrac{3}{2} \rfloor = 1.$$

Iraci–Pagaria–Paolini–Vanden Wyngaerd, Combin. Theory 3 (2023), no. 2, Paper 4; Qiu–Zhang (2026b), Section 2.2.

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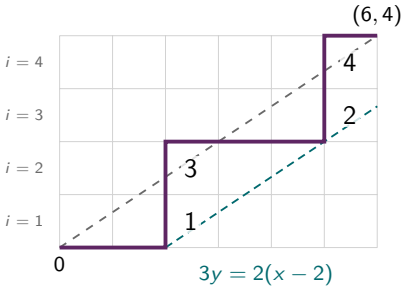
$$\mathrm{area}(\pi) = \sum_i \lfloor a_i + s \rfloor, \quad \mathrm{touch}(\pi) = \#\{i : a_i + s = 0\}.$$

Area: row 4

$$\lfloor a_4 + s \rfloor = \lfloor \tfrac{3}{2} \rfloor = 1.$$

Iraci–Pagaria–Paolini–Vanden Wyngaerd, Combin. Theory 3 (2023), no. 2, Paper 4; Qiu–Zhang (2026b), Section 2.2.

(km, kn)-preference functions: area and touches



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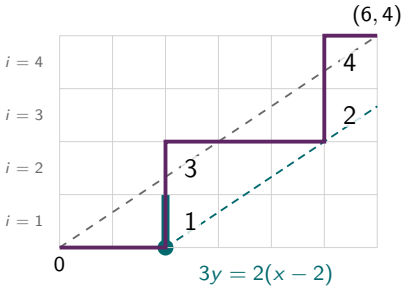
Area

$$\mathrm{area}(\pi) = 0 + 1 + 0 + 1 = 2.$$

$$\mathrm{area}(\pi) = 2,$$

Iraci–Pagaria–Paolini–Vanden Wyngaerd, Combin. Theory 3 (2023), no. 2, Paper 4; Qiu–Zhang (2026b), Section 2.2.

(km, kn)-preference functions: area and touches



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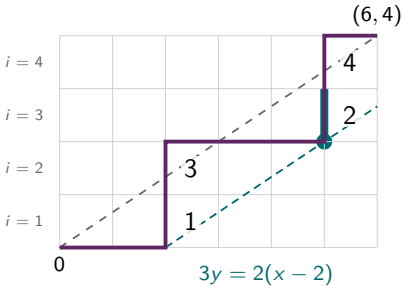
Touch: North step 1

$$a_1 + s = -2 + 2 = 0.$$

$$\mathrm{area}(\pi) = 2,$$

Iraci–Pagaria–Paolini–Vanden Wyngaerd, Combin. Theory 3 (2023), no. 2, Paper 4; Qiu–Zhang (2026b), Section 2.2.

(km, kn)-preference functions: area and touches



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The base diagonal is $my = n(x-s)$.

$$\mathrm{area}(\pi) = \sum_i \lfloor a_i + s \rfloor, \quad \mathrm{touch}(\pi) = \#\{i : a_i + s = 0\}.$$

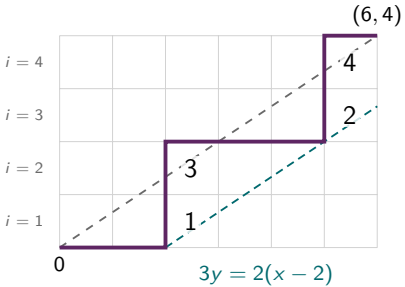
Touch: North step 3

$$a_3 + s = -2 + 2 = 0.$$

$$\mathrm{area}(\pi) = 2,$$

Iraci–Pagaria–Paolini–Vanden Wyngaerd, Combin. Theory 3 (2023), no. 2, Paper 4; Qiu–Zhang (2026b), Section 2.2.

(km, kn)-preference functions: area and touches



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The base diagonal is $my = n(x-s)$.

$$\mathrm{area}(\pi) = \sum_i \lfloor a_i + s \rfloor, \quad \mathrm{touch}(\pi) = \#\{i : a_i + s = 0\}.$$

Touches

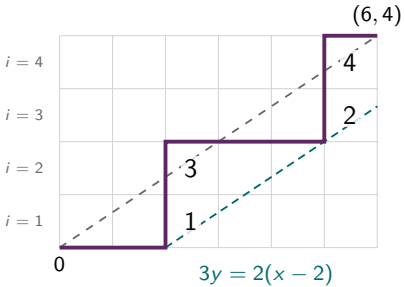
$$\{i : a_i + s = 0\} = \{1, 3\}, \qquad \mathrm{touch}(\pi) = 2.$$

$$\mathrm{area}(\pi) = 2, \qquad \mathrm{touch}(\pi) = 2.$$

Iraci–Pagaria–Paolini–Vanden Wyngaerd, Combin. Theory 3 (2023), no. 2, Paper 4; Qiu–Zhang (2026b), Section 2.2.

Rational dinv: attacking pairs

$$\mathrm{dinv}(\pi) = \underbrace{\mathrm{tdinv}(\pi)}_{} + \underbrace{\mathrm{pdinv}(\pi)}_{} - \underbrace{\mathrm{maxtdinv}(\pi)}_{} - \underbrace{B(\pi)}_{} \quad .$$



An ordered pair (i, j) is an *attack* if

$$\begin{cases} 0 \le a_j - a_i < m/n, & i < j, \\ 0 < a_j - a_i \le m/n, & i > j. \end{cases}$$

maxtdinv counts all attacks; tdinv counts attacks with $w_i < w_j$.

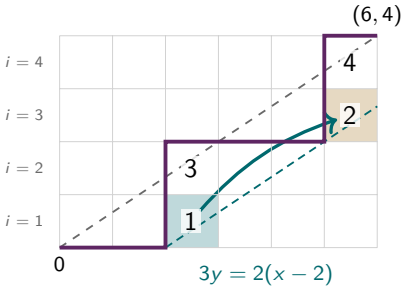
$$\mathrm{area}(\pi) = 2, \qquad \mathrm{touch}(\pi) = 2.$$

Here $m/n = 3/2$.

Qiu–Zhang (2026b), Section 2.2.

Rational dinv: attacking pairs

$$\mathrm{dinv}(\pi) = \underbrace{\mathrm{tdinv}(\pi)}_{} + \underbrace{\mathrm{pdinv}(\pi)}_{} - \underbrace{\mathrm{maxtdinv}(\pi)}_{} - \underbrace{B(\pi)}_{} \quad .$$



Pair $(1, 3)$

An ordered pair (i, j) is an *attack* if

$$\begin{cases} 0 \le a_j - a_i < m/n, & i < j, \\ 0 < a_j - a_i \le m/n, & i > j. \end{cases}$$

maxtdinv counts all attacks; tdinv counts attacks with $w_i < w_j$.

$$\mathrm{area}(\pi) = 2, \qquad \mathrm{touch}(\pi) = 2.$$

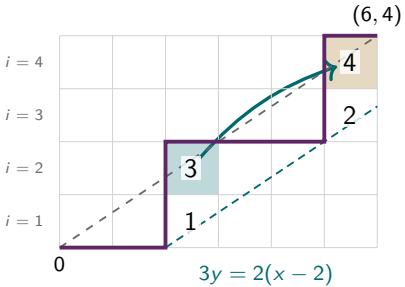
Here $m/n = 3/2$.

$$1 < 3, \quad a_3 - a_1 = 0 < \tfrac{3}{2}, \quad w_1 = 1 < w_3 = 2.$$

Qiu–Zhang (2026b), Section 2.2.

Rational dinv: attacking pairs

$$\mathrm{dinv}(\pi) = \underbrace{\mathrm{tdinv}(\pi)}_{} + \underbrace{\mathrm{pdinv}(\pi)}_{} - \underbrace{\mathrm{maxtdinv}(\pi)}_{} - \underbrace{B(\pi)}_{} \quad .$$



Pair $(2, 4)$

An ordered pair (i, j) is an *attack* if

$$\begin{cases} 0 \le a_j - a_i < m/n, & i < j, \\ 0 < a_j - a_i \le m/n, & i > j. \end{cases}$$

maxtdinv counts all attacks; tdinv counts attacks with $w_i < w_j$.

$$\mathrm{area}(\pi) = 2, \qquad \mathrm{touch}(\pi) = 2.$$

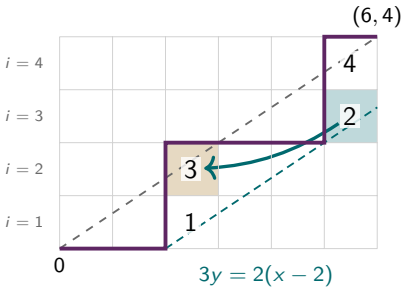
Here $m/n = 3/2$.

$$2 < 4, \quad a_4 - a_2 = 0 < \tfrac{3}{2}, \quad w_2 = 3 < w_4 = 4.$$

Qiu–Zhang (2026b), Section 2.2.

Rational dinv: attacking pairs

$$\mathrm{dinv}(\pi) = \underbrace{\mathrm{tdinv}(\pi)}_{} + \underbrace{\mathrm{pdinv}(\pi)}_{} - \underbrace{\mathrm{maxtdinv}(\pi)}_{} - \underbrace{B(\pi)}_{} \quad .$$



Pair $(3, 2)$

An ordered pair (i, j) is an *attack* if

$$\begin{cases} 0 \le a_j - a_i < m/n, & i < j, \\ 0 < a_j - a_i \le m/n, & i > j. \end{cases}$$

maxtdinv counts all attacks; tdinv counts attacks with $w_i < w_j$.

$$\mathrm{area}(\pi) = 2, \qquad \mathrm{touch}(\pi) = 2.$$

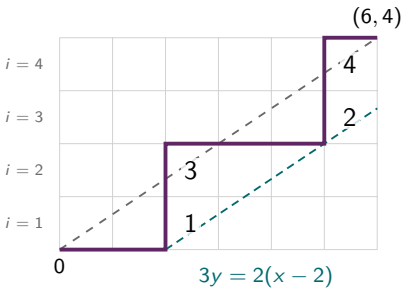
Here $m/n = 3/2$.

$$3 > 2, \quad a_2 - a_3 = \tfrac{3}{2} \le \tfrac{3}{2}, \quad w_3 = 2 < w_2 = 3.$$

Qiu–Zhang (2026b), Section 2.2.

Rational dinv: attacking pairs

$$\mathrm{dinv}(\pi) = \underbrace{\mathrm{tdinv}(\pi)}_{3} + \underbrace{\mathrm{pdinv}(\pi)}_{} - \underbrace{\mathrm{maxtdinv}(\pi)}_{3} - \underbrace{B(\pi)}_{} \quad .$$



An ordered pair (i, j) is an *attack* if

$$\begin{cases} 0 \le a_j - a_i < m/n, & i < j, \\ 0 < a_j - a_i \le m/n, & i > j. \end{cases}$$

maxtdinv counts all attacks; tdinv counts attacks with $w_i < w_j$.

$$\mathrm{area}(\pi) = 2, \qquad \mathrm{touch}(\pi) = 2.$$

Here $m/n = 3/2$.

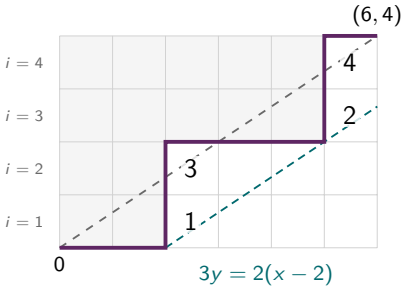
The attacks are $(1, 3), (2, 4), (3, 2)$; each has increasing labels.

$$\mathrm{tdinv}(\pi) = \mathrm{maxtdinv}(\pi) = 3.$$

Qiu–Zhang (2026b), Section 2.2.

Rational dinv: the path contribution

$$\mathrm{dinv}(\pi) = \underbrace{\mathrm{tdinv}(\pi)}_{3} + \underbrace{\mathrm{pdinv}(\pi)}_{} - \underbrace{\mathrm{maxtdinv}(\pi)}_{3} - \underbrace{B(\pi)}_{} \quad .$$



For the Ferrers diagram $\mu(\pi) = (5, 5, 2, 2)$, $\mathrm{pdinv}(\pi)$ counts cells c satisfying

$$\frac{a(c)}{\ell(c)+1} \le \frac{m}{n} < \frac{a(c)+1}{\ell(c)}.$$

Cell c; arm: right; leg: below.

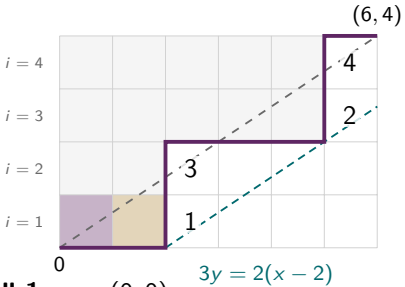
If $\ell(c) = 0$, the upper bound is $+\infty$.

$$\mathrm{area}(\pi) = 2, \qquad \mathrm{touch}(\pi) = 2.$$

Qiu–Zhang (2026b), Section 2.2.

Rational dinv: the path contribution

$$\mathrm{dinv}(\pi) = \underbrace{\mathrm{tdinv}(\pi)}_{3} + \underbrace{\mathrm{pdinv}(\pi)}_{} - \underbrace{\mathrm{maxtdinv}(\pi)}_{3} - \underbrace{B(\pi)}_{} \quad .$$



Cell 1: $c = (0, 0)$

For the Ferrers diagram $\mu(\pi) = (5, 5, 2, 2)$, $\mathrm{pdinv}(\pi)$ counts cells c satisfying

$$\frac{a(c)}{\ell(c)+1} \le \frac{m}{n} < \frac{a(c)+1}{\ell(c)}.$$

Cell c; arm: right; leg: below.

If $\ell(c) = 0$, the upper bound is $+\infty$.

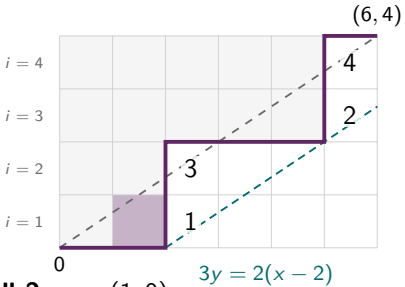
$$\mathrm{area}(\pi) = 2, \qquad \mathrm{touch}(\pi) = 2.$$

$$a(c) = 1, \quad \ell(c) = 0, \qquad \frac{1}{1} \le \frac{3}{2} < \infty.$$

Qiu–Zhang (2026b), Section 2.2.

Rational dinv: the path contribution

$$\mathrm{dinv}(\pi) = \underbrace{\mathrm{tdinv}(\pi)}_{3} + \underbrace{\mathrm{pdinv}(\pi)}_{} - \underbrace{\mathrm{maxtdinv}(\pi)}_{3} - \underbrace{B(\pi)}_{} \quad .$$



Cell 2: $c = (1, 0)$

For the Ferrers diagram $\mu(\pi) = (5, 5, 2, 2)$, $\mathrm{pdinv}(\pi)$ counts cells c satisfying

$$\frac{a(c)}{\ell(c)+1} \le \frac{m}{n} < \frac{a(c)+1}{\ell(c)}.$$

Cell c; arm: right; leg: below.

If $\ell(c) = 0$, the upper bound is $+\infty$.

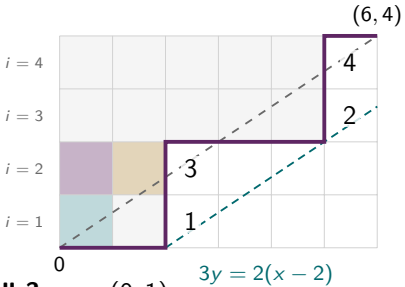
$$\mathrm{area}(\pi) = 2, \qquad \mathrm{touch}(\pi) = 2.$$

$$a(c) = 0, \quad \ell(c) = 0, \qquad \frac{0}{1} \le \frac{3}{2} < \infty.$$

Qiu–Zhang (2026b), Section 2.2.

Rational dinv: the path contribution

$$\mathrm{dinv}(\pi) = \underbrace{\mathrm{tdinv}(\pi)}_{3} + \underbrace{\mathrm{pdinv}(\pi)}_{} - \underbrace{\mathrm{maxtdinv}(\pi)}_{3} - \underbrace{B(\pi)}_{} \quad .$$



Cell 3: $c = (0, 1)$

For the Ferrers diagram $\mu(\pi) = (5, 5, 2, 2)$, $\mathrm{pdinv}(\pi)$ counts cells c satisfying

$$\frac{a(c)}{\ell(c)+1} \le \frac{m}{n} < \frac{a(c)+1}{\ell(c)}.$$

Cell c; arm: right; leg: below.

If $\ell(c) = 0$, the upper bound is $+\infty$.

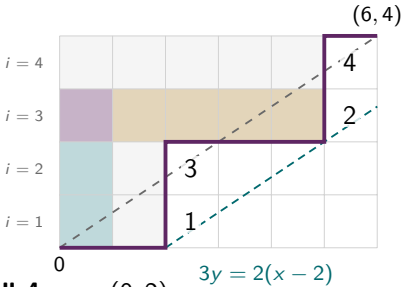
$$\mathrm{area}(\pi) = 2, \qquad \mathrm{touch}(\pi) = 2.$$

$$a(c) = 1, \quad \ell(c) = 1, \qquad \frac{1}{2} \le \frac{3}{2} < \frac{2}{1}.$$

Qiu–Zhang (2026b), Section 2.2.

Rational dinv: the path contribution

$$\mathrm{dinv}(\pi) = \underbrace{\mathrm{tdinv}(\pi)}_{3} + \underbrace{\mathrm{pdinv}(\pi)}_{} - \underbrace{\mathrm{maxtdinv}(\pi)}_{3} - \underbrace{B(\pi)}_{} \quad .$$



Cell 4: $c = (0, 2)$

For the Ferrers diagram $\mu(\pi) = (5, 5, 2, 2)$, $\mathrm{pdinv}(\pi)$ counts cells c satisfying

$$\frac{a(c)}{\ell(c)+1} \le \frac{m}{n} < \frac{a(c)+1}{\ell(c)}.$$

Cell c; arm: right; leg: below.

If $\ell(c) = 0$, the upper bound is $+\infty$.

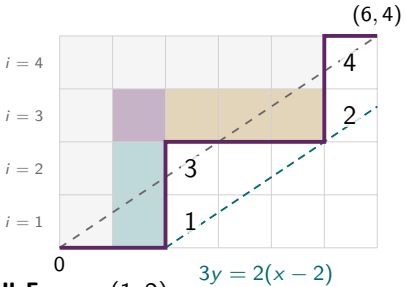
$$\mathrm{area}(\pi) = 2, \qquad \mathrm{touch}(\pi) = 2.$$

$$a(c) = 4, \quad \ell(c) = 2, \qquad \frac{4}{3} \le \frac{3}{2} < \frac{5}{2}.$$

Qiu–Zhang (2026b), Section 2.2.

Rational dinv: the path contribution

$$\mathrm{dinv}(\pi) = \underbrace{\mathrm{tdinv}(\pi)}_{3} + \underbrace{\mathrm{pdinv}(\pi)}_{} - \underbrace{\mathrm{maxtdinv}(\pi)}_{3} - \underbrace{B(\pi)}_{} \quad .$$



Cell 5: $c = (1, 2)$

For the Ferrers diagram $\mu(\pi) = (5, 5, 2, 2)$, $\mathrm{pdinv}(\pi)$ counts cells c satisfying

$$\frac{a(c)}{\ell(c)+1} \le \frac{m}{n} < \frac{a(c)+1}{\ell(c)}.$$

Cell c; arm: right; leg: below.

If $\ell(c) = 0$, the upper bound is $+\infty$.

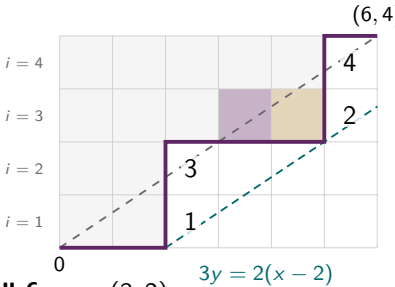
$$\mathrm{area}(\pi) = 2, \qquad \mathrm{touch}(\pi) = 2.$$

$$a(c) = 3, \quad \ell(c) = 2, \qquad \frac{3}{3} \le \frac{3}{2} < \frac{4}{2}.$$

Qiu–Zhang (2026b), Section 2.2.

Rational dinv: the path contribution

$$\mathrm{dinv}(\pi) = \underbrace{\mathrm{tdinv}(\pi)}_{3} + \underbrace{\mathrm{pdinv}(\pi)}_{} - \underbrace{\mathrm{maxtdinv}(\pi)}_{3} - \underbrace{B(\pi)}_{} \quad .$$



Cell 6: $c = (3, 2)$

For the Ferrers diagram $\mu(\pi) = (5, 5, 2, 2)$, $\mathrm{pdinv}(\pi)$ counts cells c satisfying

$$\frac{a(c)}{\ell(c)+1} \le \frac{m}{n} < \frac{a(c)+1}{\ell(c)}.$$

Cell c; arm: right; leg: below.

If $\ell(c) = 0$, the upper bound is $+\infty$.

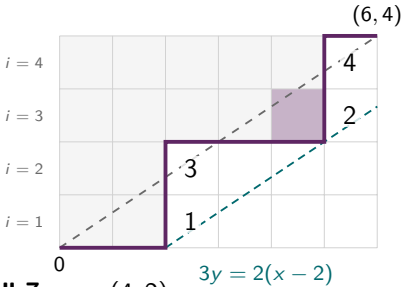
$$\mathrm{area}(\pi) = 2, \qquad \mathrm{touch}(\pi) = 2.$$

$$a(c) = 1, \quad \ell(c) = 0, \qquad \frac{1}{1} \le \frac{3}{2} < \infty.$$

Qiu–Zhang (2026b), Section 2.2.

Rational dinv: the path contribution

$$\mathrm{dinv}(\pi) = \underbrace{\mathrm{tdinv}(\pi)}_{3} + \underbrace{\mathrm{pdinv}(\pi)}_{} - \underbrace{\mathrm{maxtdinv}(\pi)}_{3} - \underbrace{B(\pi)}_{} \quad .$$



Cell 7: $c = (4, 2)$

For the Ferrers diagram $\mu(\pi) = (5, 5, 2, 2)$, $\mathrm{pdinv}(\pi)$ counts cells c satisfying

$$\frac{a(c)}{\ell(c)+1} \le \frac{m}{n} < \frac{a(c)+1}{\ell(c)}.$$

Cell c; arm: right; leg: below.

If $\ell(c) = 0$, the upper bound is $+\infty$.

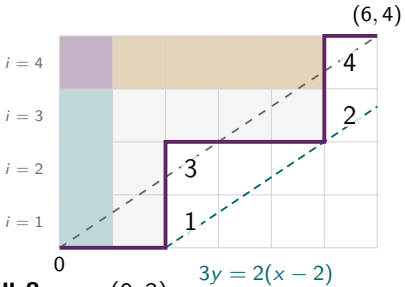
$$\mathrm{area}(\pi) = 2, \qquad \mathrm{touch}(\pi) = 2.$$

$$a(c) = 0, \quad \ell(c) = 0, \qquad \frac{0}{1} \le \frac{3}{2} < \infty.$$

Qiu–Zhang (2026b), Section 2.2.

Rational dinv: the path contribution

$$\mathrm{dinv}(\pi) = \underbrace{\mathrm{tdinv}(\pi)}_{3} + \underbrace{\mathrm{pdinv}(\pi)}_{} - \underbrace{\mathrm{maxtdinv}(\pi)}_{3} - \underbrace{B(\pi)}_{} \quad .$$



Cell 8: $c = (0, 3)$

For the Ferrers diagram $\mu(\pi) = (5, 5, 2, 2)$, $\mathrm{pdinv}(\pi)$ counts cells c satisfying

$$\frac{a(c)}{\ell(c)+1} \le \frac{m}{n} < \frac{a(c)+1}{\ell(c)}.$$

Cell c; arm: right; leg: below.

If $\ell(c) = 0$, the upper bound is $+\infty$.

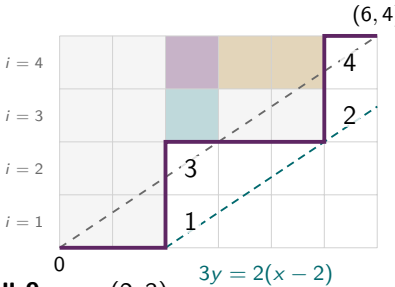
$$\mathrm{area}(\pi) = 2, \qquad \mathrm{touch}(\pi) = 2.$$

$$a(c) = 4, \quad \ell(c) = 3, \qquad \frac{4}{4} \le \frac{3}{2} < \frac{5}{3}.$$

Qiu–Zhang (2026b), Section 2.2.

Rational dinv: the path contribution

$$\mathrm{dinv}(\pi) = \underbrace{\mathrm{tdinv}(\pi)}_{3} + \underbrace{\mathrm{pdinv}(\pi)}_{} - \underbrace{\mathrm{maxtdinv}(\pi)}_{3} - \underbrace{B(\pi)}_{} \quad .$$



Cell 9: $c = (2, 3)$

For the Ferrers diagram $\mu(\pi) = (5, 5, 2, 2)$, $\mathrm{pdinv}(\pi)$ counts cells c satisfying

$$\frac{a(c)}{\ell(c)+1} \le \frac{m}{n} < \frac{a(c)+1}{\ell(c)}.$$

Cell c; arm: right; leg: below.

If $\ell(c) = 0$, the upper bound is $+\infty$.

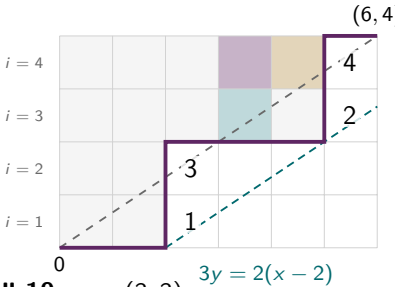
$$\mathrm{area}(\pi) = 2, \qquad \mathrm{touch}(\pi) = 2.$$

$$a(c) = 2, \quad \ell(c) = 1, \qquad \frac{2}{2} \le \frac{3}{2} < \frac{3}{1}.$$

Qiu–Zhang (2026b), Section 2.2.

Rational dinv: the path contribution

$$\mathrm{dinv}(\pi) = \underbrace{\mathrm{tdinv}(\pi)}_{3} + \underbrace{\mathrm{pdinv}(\pi)}_{} - \underbrace{\mathrm{maxtdinv}(\pi)}_{3} - \underbrace{B(\pi)}_{} \quad .$$



Cell 10: $c = (3, 3)$

For the Ferrers diagram $\mu(\pi) = (5, 5, 2, 2)$, $\mathrm{pdinv}(\pi)$ counts cells c satisfying

$$\frac{a(c)}{\ell(c)+1} \le \frac{m}{n} < \frac{a(c)+1}{\ell(c)}.$$

Cell c; arm: right; leg: below.

If $\ell(c) = 0$, the upper bound is $+\infty$.

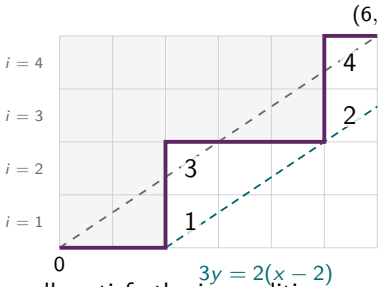
$$\mathrm{area}(\pi) = 2, \qquad \mathrm{touch}(\pi) = 2.$$

$$a(c) = 1, \quad \ell(c) = 1, \qquad \frac{1}{2} \le \frac{3}{2} < \frac{2}{1}.$$

Qiu–Zhang (2026b), Section 2.2.

Rational dinv: the path contribution

$$\mathrm{dinv}(\pi) = \underbrace{\mathrm{tdinv}(\pi)}_{3} + \underbrace{\mathrm{pdinv}(\pi)}_{10} - \underbrace{\mathrm{maxtdinv}(\pi)}_{3} - \underbrace{B(\pi)}_{} \quad .$$



Ten cells satisfy the inequalities.

For the Ferrers diagram $\mu(\pi) = (5, 5, 2, 2)$, $\mathrm{pdinv}(\pi)$ counts cells c satisfying

$$\frac{a(c)}{\ell(c)+1} \le \frac{m}{n} < \frac{a(c)+1}{\ell(c)}.$$

Cell c; arm: right; leg: below.

If $\ell(c) = 0$, the upper bound is $+\infty$.

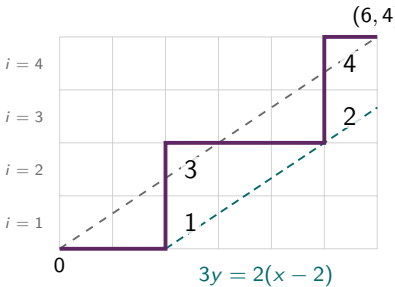
$$\mathrm{area}(\pi) = 2, \qquad \mathrm{touch}(\pi) = 2.$$

$\mathrm{pdinv}(\pi) = 10$.

Qiu–Zhang (2026b), Section 2.2.

Rational dinv: the correction below the diagonal

$$\mathrm{dinv}(\pi) = \underbrace{\mathrm{tdinv}(\pi)}_{3} + \underbrace{\mathrm{pdinv}(\pi)}_{10} - \underbrace{\mathrm{maxtdinv}(\pi)}_{3} - \underbrace{B(\pi)}_{} \quad .$$



$B(\pi)$ counts North steps whose upper endpoints lie below the main diagonal:

$$B(\pi) = \#\{i : a_i < -m/n\}.$$

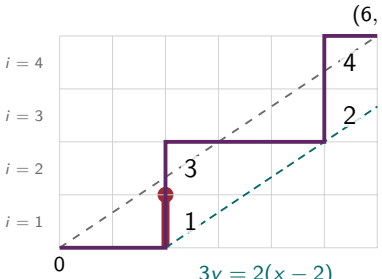
Here $m/n = 3/2$, and the main diagonal is $y = 2x/3$.

$$\mathrm{area}(\pi) = 2, \qquad \mathrm{touch}(\pi) = 2.$$

Qiu–Zhang (2026b), Section 2.2.

Rational dinv: the correction below the diagonal

$$\mathrm{dinv}(\pi) = \underbrace{\mathrm{tdinv}(\pi)}_{3} + \underbrace{\mathrm{pdinv}(\pi)}_{10} - \underbrace{\mathrm{maxtdinv}(\pi)}_{3} - \underbrace{B(\pi)}_{} \quad .$$



$B(\pi)$ counts North steps whose upper endpoints lie below the main diagonal:

$$B(\pi) = \#\{i : a_i < -m/n\}.$$

Here $m/n = 3/2$, and the main diagonal is $y = 2x/3$.

$$\mathrm{area}(\pi) = 2, \qquad \mathrm{touch}(\pi) = 2.$$

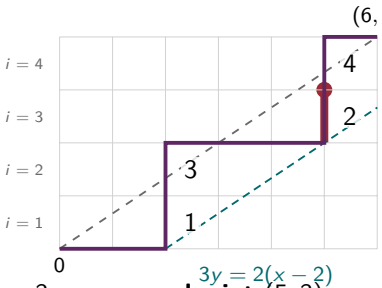
Step 1: upper endpoint $(2, 1)$

$$a_1 = -2 < -\tfrac{3}{2}, \qquad 1 < \tfrac{2}{3} \cdot 2 = \tfrac{4}{3}.$$

Qiu–Zhang (2026b), Section 2.2.

Rational dinv: the correction below the diagonal

$$\mathrm{dinv}(\pi) = \underbrace{\mathrm{tdinv}(\pi)}_{3} + \underbrace{\mathrm{pdinv}(\pi)}_{10} - \underbrace{\mathrm{maxtdinv}(\pi)}_{3} - \underbrace{B(\pi)}_{} \quad .$$



Step 3: upper endpoint $(5, 3)$

$B(\pi)$ counts North steps whose upper endpoints lie below the main diagonal:

$$B(\pi) = \#\{i : a_i < -m/n\}.$$

Here $m/n = 3/2$, and the main diagonal is $y = 2x/3$.

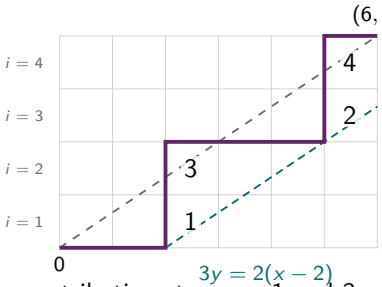
$$\mathrm{area}(\pi) = 2, \qquad \mathrm{touch}(\pi) = 2.$$

$$a_3 = -2 < -\tfrac{3}{2}, \qquad 3 < \tfrac{2}{3} \cdot 5 = \tfrac{10}{3}.$$

Qiu–Zhang (2026b), Section 2.2.

Rational dinv: the correction below the diagonal

$$\mathrm{dinv}(\pi) = \underbrace{\mathrm{tdinv}(\pi)}_{3} + \underbrace{\mathrm{pdinv}(\pi)}_{10} - \underbrace{\mathrm{maxtdinv}(\pi)}_{3} - \underbrace{B(\pi)}_{2} = 8.$$



$B(\pi)$ counts North steps whose upper endpoints lie below the main diagonal:

$$B(\pi) = \#\{i : a_i < -m/n\}.$$

Here $m/n = 3/2$, and the main diagonal is $y = 2x/3$.

$$\mathrm{area}(\pi) = 2, \qquad \mathrm{touch}(\pi) = 2.$$

The contributing steps are 1 and 3, so $B(\pi) = 2$.

$$\mathrm{dinv}(\pi) = 3 + 10 - 3 - 2 = 8.$$

Qiu–Zhang (2026b), Section 2.2.

The C-expansion of a power sum

Write $C_\alpha[X;q] = C_\alpha(1)$. For $1 \le \ell \le k$, set

$$E_{k,\ell}[X] = \sum_{\substack{\alpha \vDash k \\ \ell(\alpha)=\ell}} C_\alpha[X;q].$$

Proposition (Can–Loehr; Haglund–Morse–Zabrocki)

$$(-1)^{k-1} p_k = \sum_{\ell=1}^{k} \frac{[k]_q}{[\ell]_q} E_{k,\ell} = \sum_{\alpha \vDash k} \frac{[k]_q}{[\ell(\alpha)]_q} C_\alpha[X;q].$$

Can–Loehr, J. Combin. Theory Ser. A 113 (2006), 1419–1434; HMZ, Canad. J. Math. 64 (2012), 822–844.

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Example: $k = 2$

$$-p_2 = (1+q)C_{(2)}(1) + C_{(1,1)}(1).$$

Apply $f \longmapsto f[-MX^{m,n}] \cdot 1$.

Can–Loehr, J. Combin. Theory Ser. A 113 (2006), 1419–1434; HMZ, Canad. J. Math. 64 (2012), 822–844.

Mellit's compositional shuffle theorem

Diagonal returns: $(j_a m, j_a n)$, where $0 = j_0 < \cdots < j_\ell = k$.

$$\mathrm{comp}(\pi) = (j_1 - j_0, \ldots, j_\ell - j_{\ell-1}) \vDash k, \qquad \mathrm{touch}(\pi) = \ell.$$

Theorem (Mellit)

For every composition $\alpha \vDash k$,

$$C_\alpha[-MX^{m,n}] \cdot 1 = \sum_{\substack{\pi \in \mathrm{PF}_{km,kn} \\ \mathrm{comp}(\pi) = \alpha}} q^{\mathrm{dinv}(\pi)} t^{\mathrm{area}(\pi)} x_\pi.$$

Mellit, Duke Math. J. 170 (2021), 4123–4169; Qiu–Zhang (2026b), Theorem 2.7.

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Sum over compositions of length ℓ

$$E_{k,\ell}[-MX^{m,n}] \cdot 1 = \sum_{\substack{\pi \in \mathrm{PF}_{km,kn} \\ \mathrm{touch}(\pi) = \ell}} q^{\mathrm{dinv}(\pi)} t^{\mathrm{area}(\pi)} x_\pi.$$

Mellit, Duke Math. J. 170 (2021), 4123–4169; Qiu–Zhang (2026b), Theorem 2.7. Theorem 4.1.

A refinement by the number of touches

Theorem (Qiu–Zhang)

$$\frac{[km]_q}{[\ell]_q} E_{k,\ell}[-MX^{m,n}] \cdot 1 = \sum_{\substack{\pi \in \mathrm{Pr}_{km,kn} \\ \mathrm{touch}(\pi)=\ell}} q^{\mathrm{dinv}(\pi)} t^{\mathrm{area}(\pi)} x_\pi \qquad (1 \le \ell \le k).$$

Equivalent form via Mellit's theorem

$$[\ell]_q \sum_{\substack{\pi \in \mathrm{Pr}_{km,kn} \\ \mathrm{touch}(\pi)=\ell}} q^{\mathrm{dinv}(\pi)} t^{\mathrm{area}(\pi)} x_\pi = [km]_q \sum_{\substack{\pi \in \mathrm{PF}_{km,kn} \\ \mathrm{touch}(\pi)=\ell}} q^{\mathrm{dinv}(\pi)} t^{\mathrm{area}(\pi)} x_\pi.$$

Qiu–Zhang (2026b), Theorem 4.2 and equation (12).

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1. Paths $\longleftrightarrow$ row strict tableaux $\longleftrightarrow$ closed walks on states.
2. Fix letter multiplicities; vary the initial state.
3. Preserve area and the label multiset; compare dinv.

Qiu–Zhang (2026b), Theorem 4.2 and equation (12). Proposition 4.9.

Proof of **Theorem D**: summing over touches

Proposition (Can–Loehr; Haglund–Morse–Zabrocki)

$$(-1)^{k-1}\frac{[km]_q}{[k]_q}p_k[-MX^{m,n}]\cdot 1 = \sum_{\ell=1}^{k}\frac{[km]_q}{[\ell]_q}E_{k,\ell}[-MX^{m,n}]\cdot 1.$$

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Can–Loehr, J. Combin. Theory Ser. A 113 (2006); HMZ, Canad. J. Math. 64 (2012). Qiu–Zhang (2026b), Theorem 4.2 and proof of Theorem 4.3.

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Theorem D (Qiu–Zhang)

$$(-1)^{k-1}\frac{[km]_q}{[k]_q}p_k[-MX^{m,n}]\cdot 1=\sum_{\pi\in \mathrm{Pr}_{km,kn}} q^{\mathrm{dinv}(\pi)}t^{\mathrm{area}(\pi)}x_\pi.$$

Can–Loehr, J. Combin. Theory Ser. A 113 (2006); HMZ, Canad. J. Math. 64 (2012). Qiu–Zhang (2026b), Theorem 4.2 and proof of Theorem 4.3.

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